Design Principle of a Dual-Actuated Robotic Hand With Anthropomorphic Self-Adaptive Grasping and Dexterous Manipulation Abilities

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***Abstract*—How to use few actuators to endow a multifingered robotic hand with the human grasping and manipulation abilities is a challenging problem. This article develops a design principle of a five-fingered robotic hand with only two actuators, which guarantees the designed hand can replicate a large portion of the grasping and a limited portion of the manipulation abilities of the human hand: simultaneously move all joints before contacting objects like the natural reach-to-grasp process of the human hand; self-adaptively pinch/envelop objects with arbitrary shapes; and carry out dexterous manipulation, such as in-hand manipulation of a ball and screwing bottle cap. The core design ideas involve two levels. The higher level idea is to construct a robotic design theory ensuring thumb and fingers driven independently and four fingers driven synergistically so that the opposable thumb with manual abduction/adduction function can move coordinately with four fingers to ensure abundant anthropomorphic grasps and in-hand manipulations. The lower level idea is to establish a mechanism design method that embeds human two primary eigengrasps into a novel single-actuated differential palm mechanism consisting of gears, drag axes, and slider elements, so that the mechanism can always replicate the grasping behaviors of human hand, no matter how the differential processes it would perform during grasping. Finally, we design a mini X-hand to verify our proposed design principle and reproduce 29 types out of 33 grasping types and some manipulations that are commonly used by human hands in daily life.**

***Index Terms*—Anthropomorphism, manipulation, robotic hand, self-adaptive grasping, underactuated differential mechanism.**

I. Introduction

ONE central challenge in robotic design is to create a robotic hand with human grasping functions, dexterous manipulation abilities, and compact structure as a human hand [1]–[3]. It carries people's expectation for re-engineering the dexterity of human hands, as well as the pressing needs in automated factories, service robots, dangerous environments and rehabilitation engineering. To address this challenge, since 1980s, scientists have tried to develop fully actuated multifinger dexterous hands [4]–[6], with the expectation of actively controlling each joint to achieve flexible grasps and dexterous manipulation. Examples of such designs include the Standford/JPL Hand with three fingers and 12 motors [4], the Okada Hand with three fingers and 11 motors [5], the Utah/MIT Hand with four fingers and 32 motors [6], the Shadow Dexterous Hand with five fingers and 36 motors [7], the DLR Hand II with five fingers and 42 motors [8], etc. This fully actuated design approach has the potential to theoretically ensure robotic hands the same dexterity as the human hands. However, usage of too many actuators imposes huge sizes and weights, complex circuits, and high costs on such hands, which limit their application in real-life scenarios. Using fewer actuators to realize anthropomorphic grasping and manipulation abilities has become the main development direction in recent years.

An adaptive underactuated mechanism, which can greatly reduce the number of motors and increase the adaptive ability of fingers to grasp objects, has become an important part of the design of underactuated robotic hand. Since the 1990s, Gosselin Group made several attempts on this aspect [9]–[11], and designed a series of underactuated grippers suitable for industrial scenes (such as Mars and Sarah series, systematical introduction refers to [11]). The emergence of these industrial grippers greatly promoted the development of adaptive underactuated mechanism. After that, new adaptive underactuated mechanisms emerged based on gears and linkages [12], tendon and pulleys [13]–[17], soft materials [18], [19], origami technology [20], and memory alloy [21] (some systematic introductions refer to [22] and [23]), providing technical bases for the design of a more challenging five-fingered robotic hand with similar sizes to a human hand. In the recent decade, a series of five-fingered robotic hands with fewer motors and adaptive grasping functions have emerged. For example, RBO Hand 2 adopts the soft finger design scheme based on silicone rubber, polyester fibers, and a polyamide scaffold, and completes more than 30 daily grasping types of human hand under the control of five pneumatic sources (each finger is driven by a pneumatic source) [18]. SmartHand adopts four-actuated scheme (one motor for thumb flexion/extension, one motor

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for thumb abduction/adduction, one motor for index finger flexion/extension and one motor for the flexion/extension of the remaining three fingers), tendon-pulley finger mechanisms and a nonback-drivable adaptive grasping mechanism, integrating motors, circuits, and actuating mechanisms in palm and achieving a series of anthropomorphic grasping functions, such as grasping a ball, pinching a card, etc. [14]. A similar four-actuated scheme is also adopted by Vanderbilt Hand [24]. The full use of underactuated differential mechanism in the design of a robot hand can reduce the number of motors to one [15], [21], [25]. However, such a design inevitably leads to a fixed pattern of motion where five fingers come together, which makes it difficult to execute the in-hand manipulation and other dexterous motions and grasps of human hands that require complex coordinated movements between thumb and fingers.

Now, the difficult point focuses on establishing a design principle with novel underactuated mechanisms, which can minimize the number of actuators and maximize the realization of abundant self-adaptive grasping functions and human-level dexterous manipulative abilities. Reasonable and ingenious addressment of the contradiction is the key to overcome the core design challenge of multifingered robotic hands over decades. Chen *et al.* [26] established a mechanical implementation principle of kinematic synergies based on two cam groups, and designed a dual-actuated anthropomorphic hand that can continuously generate anthropomorphic grasping motion and achieve a variety of different grasping modes. Recently, Santina *et al.* [27] attempted to expand the design concept of single-actuated hand driven by one adaptive synergy to forming a dual-actuated design principle based on augmented adaptive synergies, and developed a robotic hand with two actuators (drive two synergistic movement features of the robot hand, respectively) that can achieve multiple grasps and even dexterous in-hand manipulation by combining the two synergistic movement features. These dual-actuated design principles make full use of the concept of dimensionality reduction based on whole-hand synergies and can achieve a few anthropomorphic grasping functions and even in-hand manipulation ability of robotic hands. However, these design principles based on whole-hand synergies do not accord with the intrinsic grouped motion characteristics of human hands (thumb and fingers are independent), which have anatomic bases [3], [28], and thus cause functional and control gaps between real human hands and robotic hands. For example, contacts between thumb tip and four fingertips are difficult for the robotic hands but are common functions of human hands.

Recent movement science research shows that the thumb movement of the human hand is independent of the other four fingers [3], [28], and there exist movement dependencies among the four fingers [3]. The anatomical evidence of the human hand indicates that the flexors of the thumb are independent of the flexors of the four fingers, and although there are transverse tendons (Juncturae tendinae) across different fingers, these tendons only involve four fingers and thus maintain the independence of the thumb [29]. Inspired by this, in our prior work [3], we established a design principle of four-actuated robotic hand for reproducing human grasping functions: two motors are in charge of thumb abduction/adduction and flexion/extension, and the other two motors drive the movement of four fingers through a unique motion transmission mechanism, which guarantee the reproduction of 97% movement characteristics of human fingers. Is that possible to use fewer actuators to still replicate the human grasping characteristics as well as dexterous manipulation capabilities? It has remarkable significance for the prosthetic hand design that requires strict sizes and as few control interfaces as possible. Since the four-actuated scheme has already carefully considered the number of actuators, it is much more difficult to further reduce the number of actuators while still reproducing the human grasping characteristics and possessing dexterous manipulation abilities. At a higher level, how to establish such a fewer-actuated principle to ensure well grasping and manipulation potentials? At a lower level, how to design new motion transmission mechanisms that can ensure the anthropomorphic self-adaptive grasping and dexterous manipulation abilities with the minimum number of motors?

Addressing these issues, this article establishes a new design principle of dual-actuated robotic hand (the thumb and four fingers are driven independently, partially inheriting our prior work [3]), which can realize abundant anthropomorphic self-adaptive grasps and some dexterous manipulations. We introduce unique differential elements consisted of gears, drag axes, and slider elements into the design of a dual-input motion transmission mechanism that can replicate the 97% grasping movement characteristics of fingers, and invent a new single-actuated differential palm mechanism. Such a mechanism can ensure anthropomorphic grasping behaviors and postures no matter how the differential process it would perform. In the way of a hybrid actuating scheme, thumb flexion/extension is actively controlled by a motor, and abduction/adduction is manually adjusted; thus, the main functions of thumb can be retained. Based on this design principle, we develop a compact dual-actuated mini X-hand with anthropomorphic self-adaptive grasping and manipulation abilities (see Fig. 1).

Three original contributions of the article include the following.

1) This article for the first time demonstrates with experiments that only two actuators with a manual DOF can ensure a robotic hand replicating about ninety percent of the human grasping activities of daily living (implementation of 29 out of 33 common grasping types of human hand in Feix taxonomy [30]). At the same time, such a hand possesses dexterous manipulation abilities, such as in-hand manipulation, screwing bottle cap, etc., and ensures anthropomorphic motion abilities in executing typically self-adaptive grasps, illustrated by high correlation between motion series of a human hand and motion series of the robotic hand.
2) This article proposes a design method of single-actuated differential mechanism, which can ensure that the designed mechanism is compact enough to be integrated in human-sized palm and can always replicate the grasping behaviors and postures of human hands, no matter how the differential processes it would perform during grasping. This is different from existing design methods of underactuated differential mechanisms that purely aim to adapt

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Fig. 1. Structure (middle column) and representative functions (three grasping examples and one in-hand manipulation example in the left column) of the dual-actuated mini X-hand. It also possesses human-like self-adaptive characteristics (one example in the right column). More functions refer to the supplementary video, where the dual-actuated design principle is demonstrated that can ensure abundant anthropomorphic self-adaptive grasping and dexterous manipulation abilities.

to object shapes. This design thought paves a new way to underactuated mechanism design for anthropomorphic robotic hand.

3) Different from the existed design methods based on whole-hand motion synergies, the proposed dual-actuated principle in this article considers thumb and fingers independently and four fingers synergistically. Such a consideration makes the designed hand carry the inherent movement characteristics of human hands [3], [28]. Such a design principle is also the basic reason why such a few-actuated design can implement a large portion of the grasping and a limited portion of the manipulation abilities of the human hand.

In the following, after discussing motion characteristics of human hands in Section II, we propose the dual-actuated design principle able to carry the movement characteristics in Section III. Quasi-static analysis of the dual-actuated grasping/manipulation properties is introduced in Section IV, and at an engineering level, implementation and test of the mini X-hand is further carried out in Section V. Finally, Section VI concludes this article.

II. Motion Characteristics of Human Hands

Human hand motion characteristics in the grasping activities of daily living can be summarized from the following three aspects: movement relationship among joints in a digit, among digits in the hand, and postural synergic characteristic of the fingers [3]. The conclusions and part of the contents of the following two paragraphs are extracted from our prior work [3].

In a digit, the ℓ-joint motion of every digit can be decomposed into ℓ simple movements along ℓ orthogonal eigen directions ${}^i\mathbf{p}_1 \in \Re^{\ell\times 1}$ to ${}^i\mathbf{p}_\ell \in \Re^{\ell\times 1}$ through principal component analysis ($i = 1, \ldots, 5$ representing index finger, middle finger, ring finger, little finger, and thumb in sequence), whereas the primary motion of the digit along the first eigen direction ${}^i\mathbf{p}_1$ mainly corresponds to the grasping pattern of the digit. The secondary motion of the digit in the space spanned by the remaining eigen directions ${}^i\mathbf{p}_2$ to ${}^i\mathbf{p}_\ell$ form the adaptive ability of the digit to the shape of the object to be grasped.

The hierarchical clustering analysis in the movement relationship among digits in the hand shows that the thumb moves independently of other four fingers in the grasping activities of daily living, and there exist movement dependencies among the four fingers. The movement dependencies imply the existence of the postural synergies or eigengrasps in the four fingers, which are verified by using the singular value decomposition to a motion dataset related to four metacarpophalangeal (MCP) joints of fingers. Two postural synergies $\mathbf{u}_1 \in \Re^{4\times 1}$ and $\mathbf{u}_2 \in \Re^{4\times 1}$ account for 97% (90% and 7%, respectively) data variances, which means that theoretically we can reconstruct complicated grasp postures by using the linear combination of the first eigengrasp [reflects approximately simultaneous flexion of four fingers, as shown in Fig. 2(a)] and the second eigengrasp (reflects medial-to-center or lateral-to-center flexion of fingers).

That is to say, human hand motions $\mathbf{Q}^{\text{hand}} \in \Re^{(5\ell+1)\times T}$ (5ℓ flexion/extension joints, 1 abduction/adduction joint of thumb, and T samples) can be constructed by primary motions of four fingers $\mathbf{Q}^{\text{fingers}}_{\text{primary}} \in \Re^{4\ell\times T}$, primary motion of thumb $\mathbf{Q}^{\text{thumb}}_{\text{primary}} \in \Re^{(\ell+1)\times T}$ and secondary motion of each digit ${}^i\mathbf{q}_{\text{secondary}} \in \Re^{\ell\times T}$ formed by ${}^i\mathbf{p}_2$ to ${}^i\mathbf{p}_\ell$ and corresponding actuating signals ${}^i\mathbf{l}_2 \in \Re^{1\times T}$ to ${}^i\mathbf{l}_\ell \in \Re^{1\times T}$ [see (1) and (2)]. The 4ℓ-dimensional primary motion series of four fingers can be generated by only two-dimensional (2-D) motions $[\,{\sigma_1}^T\ {\sigma_2}^T\,]^T \in \Re^{2\times T}$ through a synergic motion transmission relation as shown in (2). The $(\ell+1)$-dimensional primary motion of thumb can be represented by two independent motions $\sigma_3 \in \Re^{1\times T}$ in charge of thumb flexion/extension and $\sigma_4 \in \Re^{1\times T}$in charge of thumb abduction/adduction, as shown in (2)

$$\mathbf{Q}^{\text{hand}} = \begin{bmatrix} \mathbf{Q}^{\text{fingers}}_{\text{primary}} \\ \mathbf{Q}^{\text{thumb}}_{\text{primary}} \end{bmatrix} + \begin{bmatrix} {}^1\mathbf{q}_{\text{secondary}}{}^T \cdots {}^5\mathbf{q}_{\text{secondary}}{}^T\ \mathbf{0} \end{bmatrix}^T \tag{1}$$

where

$$\begin{cases} \mathbf{Q}^{\text{fingers}}_{\text{primary}} = \text{blkdiag}\left\{ {}^1\mathbf{p}_1/{}^1p_{11}, \ldots, {}^4\mathbf{p}_1/{}^4p_{11} \right\} \begin{bmatrix} \mathbf{u}_1\, \mathbf{u}_2 \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix} \\ \mathbf{Q}^{\text{thumb}}_{\text{primary}} = \text{blkdiag}\left\{ {}^5\mathbf{p}_1/{}^5p_{11}, 1 \right\} \begin{bmatrix} {\sigma_3}^T\ {\sigma_4}^T \end{bmatrix}^T. \\ {}^i\mathbf{q}_{\text{secondary}} = \sum_{j=2}^{\ell} {}^i\mathbf{p}_j\, {}^i\mathbf{l}_j, i = 1, \ldots, 5 \end{cases} \tag{2}$$

Note that ${}^ip_{11}$ is the first element of ${}^i\mathbf{p}_1$ ($i = 1, \ldots, 5$).

III. Dual-Actuated Design Principle

Note that, single-actuated hands usually have a fixed movement form that limits the ability of dexterous manipulation. The dual-actuated scheme is considered as the fewest-actuated

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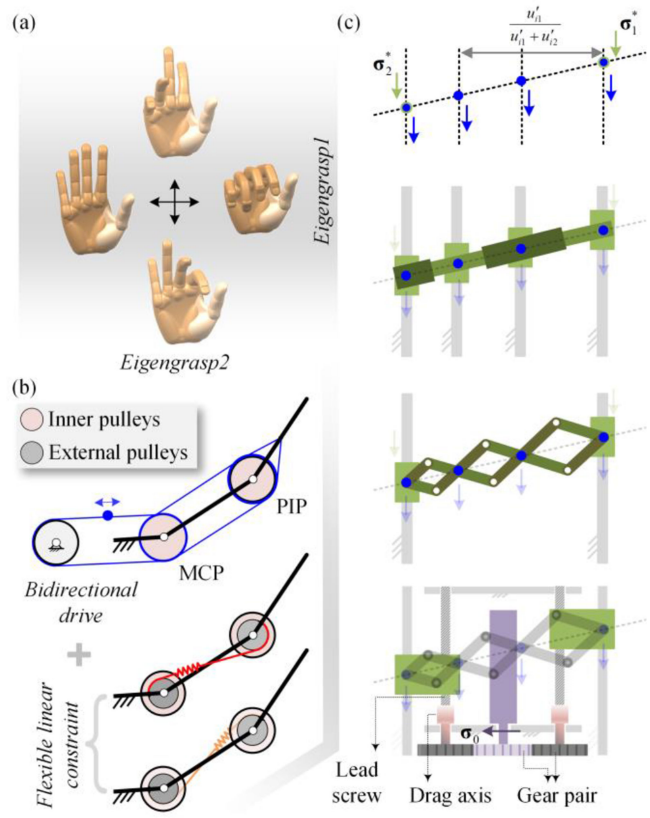


Fig. 2. Design of single-actuated differential mechanism for reproducing human grasping characteristics among four fingers. (a) Two eigengrasps, which can represent human grasping characteristics among four fingers. (b) Design principle of a single-actuated two-phalange mechanical digit able to implement primary motion and secondary motion among joints inside a human digit. (c) Toward the single-actuated differential mechanism for reproducing human grasping characteristics constructed by the two eigengrasps in (a). The first picture shows geometrical relationships of points that can implement the grasping motion characteristics among four fingers: all points must keep colinear and paths on which the points move must be parallel. The slider-based mechanism is a direct dual-actuated mechanism scheme that implements the geometrical relationship (second picture). It can be evolved to a scissor-like-element-based mechanism, which needs less guiders (third picture). Introducing drag axes and gears, the dual-actuated scissor-like-element-based mechanism can be further evolved to the final single-actuated differential mechanism.

scheme for achieving both flexible grasping and manipulation. While using four actuators to match the four inputs σ_1, σ_2, σ_3, and σ_4 is a natural choice, is that possible to mechanically implement the movement characteristics in an only dual-actuated framework?

In this section, we discuss how to construct the dual-actuated hand design principle, so as to endow the dual-actuated anthropomorphic hand with human hand motion characteristics described by (2) in the last section.

A. Design of a Single-Actuated Two-Phalange Mechanical Digit for Reproducing the Primary and Secondary Motions

Despite that a human digit has three phalanges, here we simplify mechanical digit design to two phalanges, which is widely adopted in current prosthetic hand design, such as the SSSA-MyHand [31] and most of the current commercial prosthetic hands. We define $\ell = 2$. According to the motion characteristics in a human digit, the digit design needs to ensure a linear motion relationship of joints before contacting an object (implement the primary motion), and self-adaptively grasping object after contacting object (implement the secondary motion). However, it is a great challenge to embed the natural flexion/extension characteristics as well as grasping compliant characteristics into a compact mechanical digit in the form of mechanical intelligence [32].

Inheriting our previous work [3], [33], an underactuated two-phalange digit design scheme consisted of springs, pulleys, and cyclically driven tendon [refer to Figs. 2(b) and 3(a)] is adopted. It can ensure the following:

1) the compactness of the digit structure;
2) bidirectional drive;
3) the linear constraint relation of the motion of each joint during natural flexion and extension (see following);
4) the compliant grasping ability.

A pair of springs on two sides of the digit reduces the 2-DOF finger to the single-DOF finger in the way of flexible constraint, which ensures that each joint of the digit moves simultaneously before touching the object, and the motion of the two joints presents an inherent linear constraint relationship (implement the primary motion, the relationship is introduced in the following). After touching the object, due to the contact forces exerting on the phalanges, one of the springs in the pair of springs will be extended, which breaks the linear constraint in the natural motions and ensure that the digit can comply with the shape of the object (implement the secondary motion). The cyclically driven tendon inside the digit ensures that the rope tension is applied to each joint, and the finger's flexion and extension can be actively controlled by the motor, ensuring good grasping and manipulation potentials of the digit.

We define ${}^{i}r_1$, ${}^{i}r_2$ as the inner pulley radii of the MCP joint and proximal interphalangeal (PIP) joint of the *i*th digit, ${}^{i}r'_1$ and ${}^{i}r'_2$ the corresponding external pulley radii, and ${}^{i}k$ the spring stiffness of the *i*th digit [as shown in Fig. 3(a)]. The motion transmission relationship between the MCP joint angle ${}^{i}\mathbf{q}_1 \in \Re^{1\times T}$, PIP joint angle ${}^{i}\mathbf{q}_2 \in \Re^{1\times T}$, displacement of the tendon ${}^{i}\mathbf{l} \in \Re^{1\times T}$, and deformation of the spring ${}^{i}\mathbf{x} \in \Re^{1\times T}$ can be written as

$$\begin{bmatrix} {}^{i}\mathbf{q}_1 \\ {}^{i}\mathbf{q}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} {}^{i}r_1 & {}^{i}r_2 \\ -{}^{i}r'_1 & {}^{i}r'_2 \end{bmatrix}^{-1}}_{{}^{i}\mathbf{M}\in\Re^{2\times 2}} \begin{bmatrix} {}^{i}\mathbf{l} \\ {}^{i}\mathbf{x} \end{bmatrix}. \tag{3}$$

The above relationship indicates that in natural motion, the springs are theoretically free from deformation (i.e., ${}^{i}\mathbf{x} = \mathbf{0}$), and there is an inherent simultaneous movement relationship and linear constraint relationship between joints: ${}^{i}r'_2{}^{i}\mathbf{q}_2 - {}^{i}r'_1{}^{i}\mathbf{q}_1 = \mathbf{0}$. If the pulley radii satisfy ${}^{i}r'_1/{}^{i}r'_2 = {}^{i}p_{11}/{}^{i}p_{12}$, the preshaping motions of the mechanical digit would conform to the primary motion of a human digit. After contacting objects, the linear proportional relationship can be broken due to the deformation of the springs, so that the secondary motion of a human digit can then be implemented on the mechanical digit.

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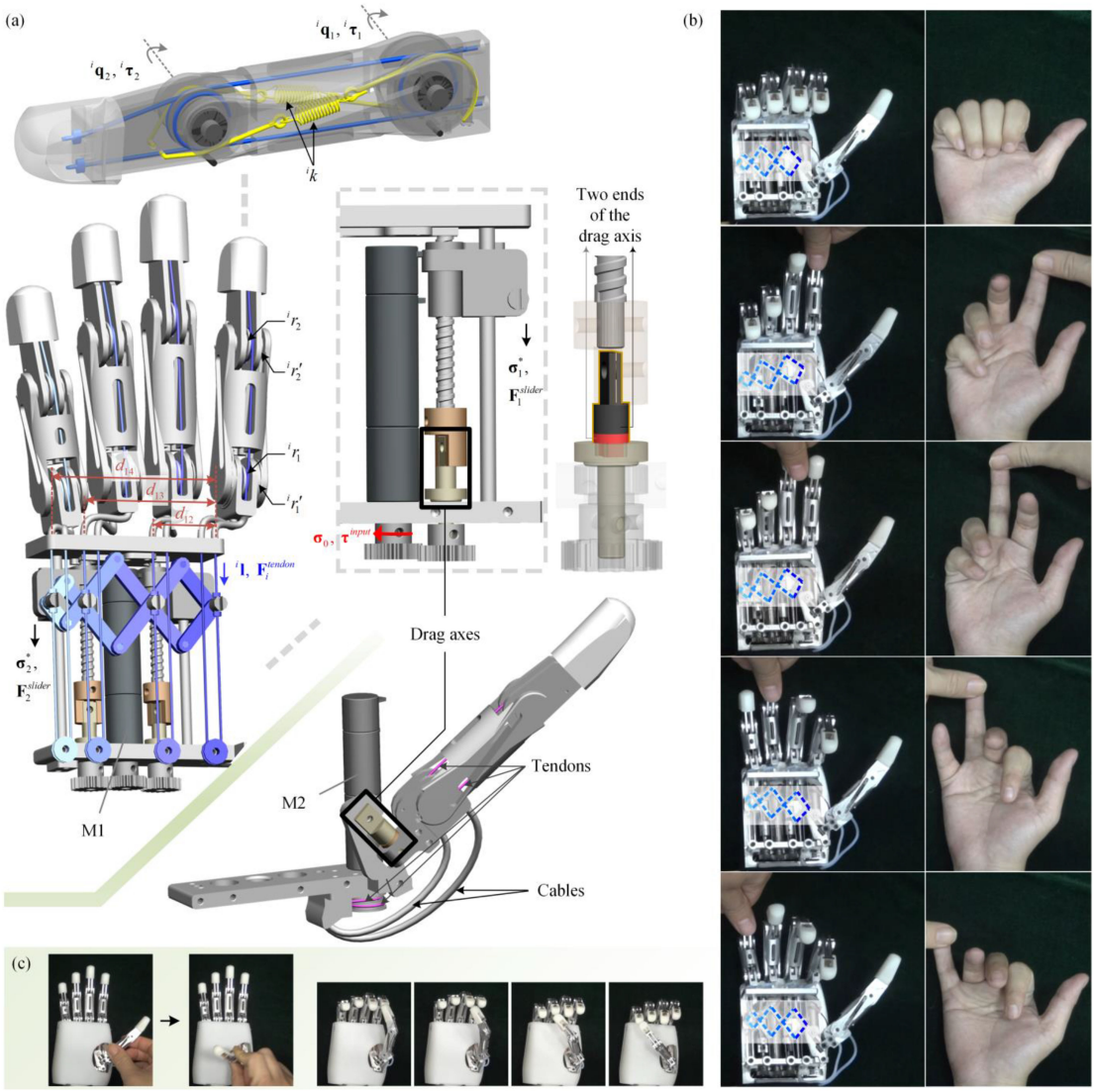


Fig. 3. Dual-actuated design principle. (a) Overall design principle, consisting of the differential mechanism inside a finger and among the four fingers, and the manually adjustable mechanism of the thumb. (b) Underactuated differential mechanism ensures self-adaptive characteristics of human fingers. (c) Thumb design ensures wide abduction/adduction range and thumb-finger opposition.

In addition, according to the dual principle, the relationship among the joint torques ${}^i\tau_1 \in \Re^{1\times T}$, ${}^i\tau_2 \in \Re^{1\times T}$, the tendon force $\mathbf{F}_i^{\text{tendon}} \in \Re^{1\times T}$, and the tension of the spring ${}^ik^i\mathbf{x}$ is

$$\begin{bmatrix} \mathbf{F}_i^{\text{tendon}} \\ {}^ik^i\mathbf{x} \end{bmatrix} = {}^i\mathbf{M}^T \begin{bmatrix} {}^i\tau_1 \\ {}^i\tau_2 \end{bmatrix}. \tag{4}$$

B. Actuating Module of Fingers: Design of Single-Actuated Differential Mechanism for Reproducing Human Grasping Characteristics Among Fingers

To replicate the grasping motion characteristics among four fingers, dual-actuated motion transmission mechanism is normally needed to mechanically reconstruct the motion plane spanned by the two eigengrasps $\mathbf{u}_1$ and $\mathbf{u}_2$, as shown in Fig. 2(a). To ensure the motion transmission relation $\begin{bmatrix}\mathbf{u}_1 & \mathbf{u}_2\end{bmatrix}\begin{bmatrix}\sigma_1 \\ \sigma_2\end{bmatrix}$, two-way actuating signals $\begin{bmatrix}\sigma_1^* \\ \sigma_2^*\end{bmatrix} \in \Re^{2\times T}$ can be set as $\begin{bmatrix}\sigma_1^* \\ \sigma_2^*\end{bmatrix} = \begin{bmatrix}c_1 & \\ & c_4\end{bmatrix}\begin{bmatrix}u_{11} & u_{12} \\ u_{41} & u_{42}\end{bmatrix}\begin{bmatrix}\sigma_1 \\ \sigma_2\end{bmatrix}$ (where c_1, c_4 are constants to ensure practical meanings of the actuating signals) so that motion transmission relation $\begin{bmatrix}\mathbf{u'}_1 & \mathbf{u'}_2\end{bmatrix} \in \Re^{4\times 2}$ between four MCP joints and 2-D motion inputs is $\begin{bmatrix}\mathbf{u'}_1 & \mathbf{u'}_2\end{bmatrix} = \begin{bmatrix}\mathbf{u}_1 & \mathbf{u}_2\end{bmatrix}\begin{bmatrix}u_{11} & u_{12} \\ u_{41} & u_{42}\end{bmatrix}^{-1}\begin{bmatrix}c_1 & \\ & c_4\end{bmatrix}^{-1} = \begin{bmatrix}{c_1}^{-1} & u'_{21} & u'_{31} & 0 \\ 0 & u'_{22} & u'_{32} & {c_4}^{-1}\end{bmatrix}^T$. Since the general motion transmission matrix $\begin{bmatrix}\mathbf{u'}_1 & \mathbf{u'}_2\end{bmatrix}$ can be always decomposed into a 4×4diagonal matrix $\text{diag}\{{c_1}^{-1}, u'_{21}+u'_{22}, u'_{31}+u'_{32}, {c_4}^{-1}\}$ in which each diagonal element can be implemented by adjusting pulley radii in a mechanical finger [through satisfying constraint (5) obtained based on (2) and (3)], and a motion transmission matrix $\begin{bmatrix}1 & u'_{21}/(u'_{21}+u'_{22}) & u'_{31}/(u'_{31}+u'_{32}) & 0 \\ 0 & u'_{22}/(u'_{21}+u'_{22}) & u'_{32}/(u'_{31}+u'_{32}) & 1\end{bmatrix}^T \in \Re^{4\times 2}$

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with the property of "row sum being one," we just need to consider the mechanism design implementing the later transmission matrix. A geometrical principle in our prior work [3] indicated that if we abstract the two motion inputs and four motion outputs as points moving on parallel lines [as shown in Fig. 2(c)] and the points keep colinear during motion, they can implement the motion transmission matrix with the property of "row sum being one." The slider-based mechanism is a direct mechanism scheme that implements the geometrical relationship [3], and the mechanism can be evolved to a scissor-like-element based mechanism that needs less guiders and can also implement the relationship (as shown in the middle panels of Fig. 2(c), it will be introduced in details in another prepared paper)

$$\begin{bmatrix} {}^{i}r_1 & {}^{i}r_2 \\ -{}^{i}r'_1 & {}^{i}r'_2 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{{}^{i}\mathbf{p}_1}{{}^{i}p_{11}} \left(u'_{i1} + u'_{i2} \right), i = 1, \ldots, 4. \quad (5)$$

Is that possible to use only one actuator and still replicate the motion transmission matrix mechanically?

If we make a close watch for human natural grasping, the four fingers always move at the same time due to the existence of transverse tendons (Juncturae tendinae) that involve these fingers. If the motion of one finger is obstructed during the natural flexion process, the postures finally formed by a human hand can be found in Fig. 3(b). It is clear that if the motion of the index or middle finger (ring or little finger) is blocked, the four fingers of each hand will eventually form lateral-to-center (medial-to-center) flexion of fingers. Natural flexion of four fingers is similar to the first eigengrasp, and self-adaptive flexion of four fingers is like the second eigengrasp. This natural motion characteristic and self-adaptive characteristic of the human hand may be a comprehensive result of the extrinsic flexors (flexor digitorum profundus and flexor digitorum superficialis) that both drive all fingers simultaneously and transverse tendons (Juncturae tendinae) that connect each finger.

Inspired by the characteristics of human hands, we introduce differential elements consisting of gears and drag axes in the abovementioned dual-actuated scissor-like-element based motion transmission mechanism [as shown in Fig. 2(c)], and design a single-actuated differential mechanism that can always perform the human grasping behaviors and postures on the motion plane spanned by the two eigengrasps $\mathbf{u}_1$ and $\mathbf{u}_2$. The motor drives the rotations of the two drag axes through gear pairs. The drag axis consists of two ends that connect together through interference fit, so that they can maintain a torque when they are in the state of relative sliding [see Fig. 3(a)]. During natural motions before contacting an object, due to the small force exerted on the sliders and small torque exerted on the drag axes, no sliding happens between the two ends of the drag axes so that the two sliders can move downwards synchronously, and the four hinge pins on the scissor mechanism move downwards synchronously. The four hinge pins connect tendons that move through cables and flex/extend the four fingers. Such a Bowden-cable scheme is extremely compact, and can ensure well force transmission properties and grasping qualities, which has been verified by many existing robotic hands, such as SmartHand [14] and Vanderbilt Hand [24]. The synchronous movements of the four hinge pins lead the simultaneous movements of the tendons and the simultaneous flexion of the four fingers, which is similar to the natural flexion of a human hand. After contacting an object and one slider is subjected to sufficient resistance, the corresponding lead screw and its associated drag axis (one end) stops rotating, whereas the other end of the drag axis (which connects the gear to the motor) can continue to rotate and maintain a fixed torque $\tau_{\max}^{\text{dragaxis}}$. Therefore, the motor can still control the other slider to move continually, and the scissor mechanism deforms, and differential movements are formed between the two sliders. The synchronous flexions of the four fingers are changed to medial-to-center flexion or lateral-to-center flexion so as to adapt to the shape of the objects. It can be seen from the supplementary video and Fig. 3(b) that the self-adaptive characteristics of such a mechanism are very similar to those of a human hand.

Let d_{1i} $(i = 2, 3, 4)$ denote horizontal distances between the first and the *i*th tendons, the relationship between the displacements of four tendons ${}^{i}\mathbf{l} \in \Re^{1\times T}$ $(i = 1, 2, 3, 4)$ and displacements of two sliders $\sigma_j^* \in \Re^{1\times T}$ $(j = 1, 2)$ can be described as

$$\begin{bmatrix} {}^{1}\mathbf{l} \\ {}^{2}\mathbf{l} \\ {}^{3}\mathbf{l} \\ {}^{4}\mathbf{l} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 - d_{12}/d_{14} & 1 - d_{13}/d_{14} & 0 \\ 0 & d_{12}/d_{14} & d_{13}/d_{14} & 1 \end{bmatrix}^T}_{\mathbf{N}\in\Re^{4\times 2}} \begin{bmatrix} \sigma_1^* \\ \sigma_2^* \end{bmatrix}. \quad (6)$$

According to the dual principle, the relationship between the corresponding tendon forces $\mathbf{F}_i^{\text{tendon}}$ $(i = 1, 2, 3, 4)$ and actuating force of the two sliders $\mathbf{F}_j^{\text{slider}} \in \Re^{1\times T}$ $(j = 1, 2)$ is

$$\begin{bmatrix} \mathbf{F}_1^{\text{slider}} \\ \mathbf{F}_2^{\text{slider}} \end{bmatrix} = \mathbf{N}^T \begin{bmatrix} \mathbf{F}_1^{\text{tendon}} \\ \mathbf{F}_2^{\text{tendon}} \\ \mathbf{F}_3^{\text{tendon}} \\ \mathbf{F}_4^{\text{tendon}} \end{bmatrix}. \quad (7)$$

If we make $d_{12}/d_{14} = u'_{22}/(u'_{21} + u'_{22})$and $d_{13}/d_{14} = u'_{32}/(u'_{31} + u'_{32})$, the motion transmission matrix $\mathbf{N}$ implements the motion transmission matrix $\begin{bmatrix} 1 & u'_{21}/(u'_{21} + u'_{22}) & u'_{31}/(u'_{31} + u'_{32}) & 0 \\ 0 & u'_{22}/(u'_{21} + u'_{22}) & u'_{32}/(u'_{31} + u'_{32}) & 1 \end{bmatrix}^T$, which implies that the mechanism can always ensure the human grasping behaviors constructed by the two eigengrasps in Fig. 2(a), no matter how the differential processes it would perform.

We further focus on the differential part consisted of a motor, gears, drag axes, lead screws, and two sliders. Define pitch of the lead screws as p, the relative rotation angle at both ends of the *j*th drag axis as $\mathbf{c}_j$ $(j = 1, 2)$. If all gears are same, the relationship between displacements of two sliders σ_j^* $(j = 1, 2)$ and the rotation angle of the motor shaft $\sigma_0 \in \Re^{1\times T}$ can be written as

$$\begin{bmatrix} \sigma_1^* \\ \sigma_2^* \end{bmatrix} = p \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} \sigma_0 \\ \mathbf{c}_1 \\ \mathbf{c}_2 \end{bmatrix}. \quad (8)$$

The relationship between the output torque of the motor $\tau^{\text{input}} \in \Re^{1\times T}$ and the actuating force of the two sliders $\mathbf{F}_j^{\text{slider}}$ $(j = 1, 2)$ can be obtained as follows:

$$\tau^{\text{input}} = p \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{F}_1^{\text{slider}} \\ \mathbf{F}_2^{\text{slider}} \end{bmatrix}. \quad (9)$$

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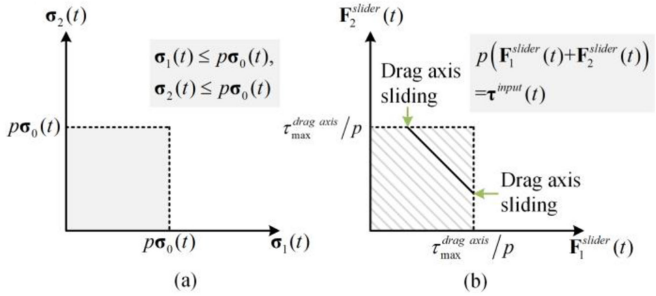


Fig. 4. Motion and force transmission relationship in the differential part.

The above relationships (8) and (9) represent the motion and force transmission relations of the differential part. When the motor shaft reaches a certain amount of rotation $\sigma_0(t)$ $(t = 1, \ldots, T)$, the displacements of the two sliders may be distributed at any point within the rectangular region formed by $\sigma_1^*(t) \le p\sigma_0(t)$ and $\sigma_2^*(t) \le p\sigma_0(t)$ [as shown in Fig. 4(a)]. When the output torque of the motor is τ^{input}, the actuating forces of the two sliders $\mathbf{F}_j^{\text{slider}}$ $(j = 1, 2)$ may be distributed at any point on a line segment in Fig. 4(b). When the actuating forces of the two sliders move to one end of the line segment due to sufficiently large external forces, the actuating forces reach the torque limit (or slipping torque) of the drag axis $\tau_{\max}^{\text{dragaxis}}$, making the two ends of a drag axis start to slip and maintain the torque.

Defining $\mathbf{n}_i$ the ith row of the motion transmission matrix $\mathbf{N}$ and connecting (3) and (6), we have

$$\begin{bmatrix} {}^i\mathbf{q}_1 \\ {}^i\mathbf{q}_2 \end{bmatrix} = {}^i\mathbf{M} \begin{bmatrix} \mathbf{n}_i & \\ & 1 \end{bmatrix} \begin{bmatrix} \begin{bmatrix} \sigma_1^* \\ \sigma_2^* \end{bmatrix} \\ {}^i\mathbf{x} \end{bmatrix} \tag{10}$$

and

$$\begin{bmatrix} \sigma_1^* \\ \sigma_2^* \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}}_{\mathbf{T'}_1 \in \Re^{2\times 2}} \begin{bmatrix} \sigma_1^* \\ \sigma_1^* - \sigma_2^* \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}}_{\mathbf{T'}_2 \in \Re^{2\times 2}} \begin{bmatrix} \sigma_1^* - \sigma_2^* \\ \sigma_2^* \end{bmatrix} \tag{11}$$

and we can reach (12), shown at the bottom of this page. This formula describes the quantitative relationship among the joint flexion amount under unit drive ${}^i\mathbf{q}_j(t)/\sigma_0(t)$, spring deformation amount under unit drive ${}^i\mathbf{x}(t)/(p\sigma_0(t))$, and the differential component under unit drive $(\sigma_1^*(t) - \sigma_2^*(t))/(p\sigma_0(t))$, describing the whole self-adaptive process. According to this formula, the self-adaptive process of the mechanism can be visualized (see Fig. 7).

C. Actuating Module of Thumb: Hybrid Drive

In order to ensure that the robotic thumb possesses the capability of a human thumb, two motors are usually needed to realize the flexion/extension and abduction/adduction ability, respectively. Here, we simplify the actuating scheme and adopt a hybrid actuating scheme: thumb flexion/extension is actively controlled by a motor, and abduction/adduction is manually adjusted by the user. This design considers the practical application of prosthetic hands, which requires extremely strict size constraints. Many current commercial prosthetic hand designs, such as Bebionic hand (Otto Bock; Duderstadt, Germany), iLimb hand (Ossur; Reykjavik, Iceland), adopt this scheme, and we continue this design thought. For other industrial applications, thumb abduction/adduction may be driven by a motor or may be frozen.

The mechanism design that implements thumb flexion/extension adopts the abovementioned digit design scheme. The thumb position design ensures good thumb-finger opposition ability [the thumb tip can touch the fingertips of four fingers in turn, as shown in Fig. 3(c)]. The good opposable ability is the base for achieving stable grasping and dexterous manipulations [34]. In order to manually adjust the thumb abduction/adduction, drag axis is utilized to connect thumb with palm. Drag axis has a certain holding torque, which can not only ensure that the thumb root rotates around its axis but also make the thumb stay in any direction, thus realizing the ability of stepless adjustment

IV. Quasi-Static Analysis of the Dual-Actuated Grasping and Manipulation

The reason that the dual-actuated design can carry the movement characteristics of human hand is to decompose human hand movements into primary and secondary movements and implement them by motors, passive elements (drag axes and springs) and a unique mechanical transmission system including both differential and coupled mechanisms. When a hand designed based on the dual-actuated principle grasps or manipulates an object, it begins with performing the human primary movements before contacting the object. After contacting the object, external disturbances (object movements, grasping forces, etc.) induce passive movements of the robotic hand, which promote the hand to implement the anthropomorphic grasping and manipulation functions. What is the determination relationship of the passive movements? Due to the dependence between the passive movements and grasping forces, are the grasping forces adjustable after the hand grasps an object? In other words, is the hand able to grasp an object in both strongly and gently ways? Is such a hand able to manipulate an object in the hand? To deepen the understanding of the dual-actuated design principle,

$$\begin{bmatrix} {}^i\mathbf{q}_1(t)/\sigma_0(t) \\ {}^i\mathbf{q}_2(t)/\sigma_0(t) \end{bmatrix} = \begin{cases} p^i\mathbf{M} \begin{bmatrix} \mathbf{n}_i & \\ & 1 \end{bmatrix} \begin{bmatrix} \mathbf{T'}_1 & \\ & 1 \end{bmatrix} \begin{bmatrix} \begin{bmatrix} 1 \\ (\sigma_1^*(t) - \sigma_2^*(t))/\max\{\sigma_1^*(t), \sigma_2^*(t)\} \end{bmatrix} \\ {}^i\mathbf{x}(t)/\max\{\sigma_1^*(t), \sigma_2^*(t)\} \end{bmatrix} & \frac{\max\{\sigma_1^*(t), \sigma_2^*(t)\}}{p\sigma_0(t)}, \sigma_1^*(t) \ge \sigma_2^*(t) \\ p^i\mathbf{M} \begin{bmatrix} \mathbf{n}_i & \\ & 1 \end{bmatrix} \begin{bmatrix} \mathbf{T'}_2 & \\ & 1 \end{bmatrix} \begin{bmatrix} \begin{bmatrix} (\sigma_1^*(t) - \sigma_2^*(t))/\max\{\sigma_1^*(t), \sigma_2^*(t)\} \\ 1 \end{bmatrix} \\ {}^i\mathbf{x}(t)/\max\{\sigma_1^*(t), \sigma_2^*(t)\} \end{bmatrix} & \frac{\max\{\sigma_1^*(t), \sigma_2^*(t)\}}{p\sigma_0(t)}, \sigma_1^*(t) < \sigma_2^*(t) \end{cases} \tag{12}$$

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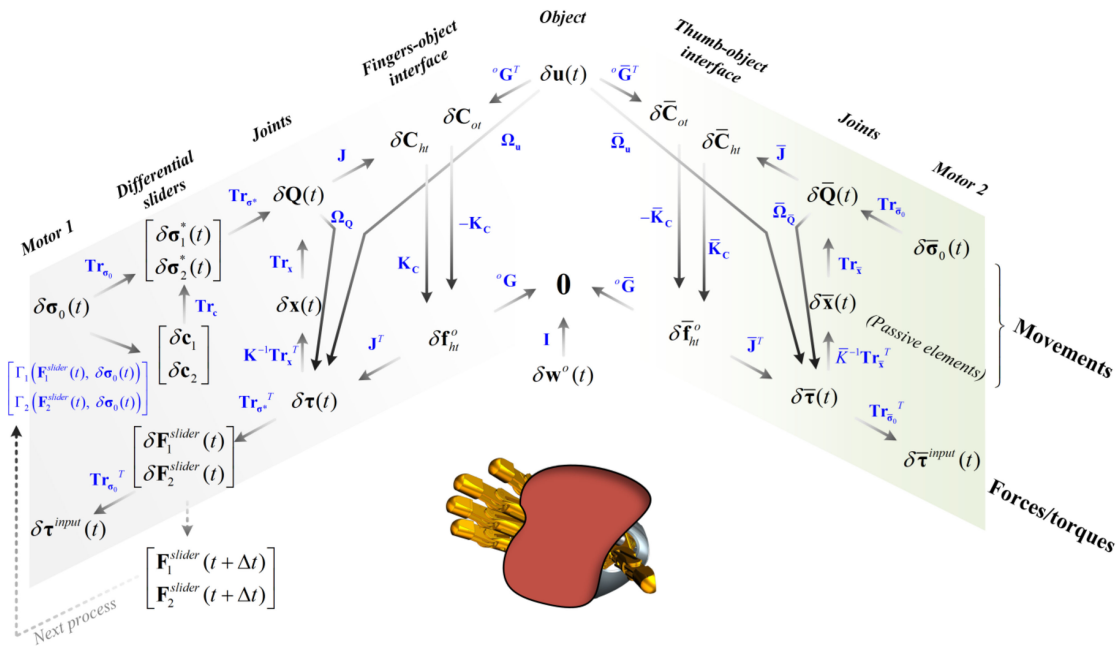


Fig. 5. Quasi-static description of the hand-object system. The blue matrices near the solid arrows indicate transmission matrices. Notations refer to Table I.

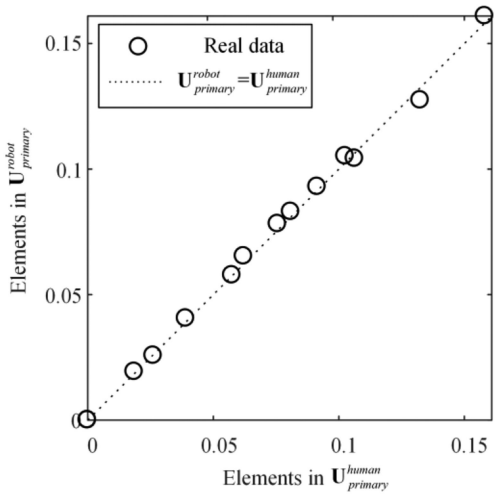


Fig. 6. Visualization of the relationship between $\mathbf{U}^{\text{human}}_{\text{primary}}$ and $\mathbf{U}^{\text{robot}}_{\text{primary}}$.

in this section, we carry out quasi-static analysis to clarify the determination factors of passive movements, the grasping force controllability and manipulability of a robotic hand designed based on the dual-actuated design principle.

A. *Quasi-Static Description of the Hand-Object System*

The hand-object system can be decomposed into fingers-object system and thumb-object system. The fingers-object system contains a motor (motor 1), differential sliders, joints, fingers/object interface, and the object. The thumb-object system contains a motor (motor 2), joints, thumb/object interface, and the object.

Before introducing the quasi-static equations, we note that for quasi-static equations and notation definition in the links of joints, hand-object interface, and the object, we refer to the book [35] and the work done by Bicchi Group [25], [36]. Detailed explanations and descriptions of some classical definitions, such as wrench, grasp matrix, contact forces, and Jacobian matrix, are not much introduced here due to the limitations of space, and can be found in these literatures. Description and visualization of the complete quasi-static process can be overseen in Fig. 5. Notations are summarized in Table I.

We start with an object equilibrium equation that establishes the relationship among external wrench acting on the object $\mathbf{w}^o(t) \in \Re^6$, contact forces by the fingers on the object $\mathbf{f}^o_{ht} \in \Re^{nc}$, and contact forces by the thumb on the object $\bar{\mathbf{f}}^o_{ht} \in \Re^{\bar{n}\bar{c}}$ through introducing the grasp matrix of fingers ${}^o\mathbf{G} \in \Re^{6\times nc}$ and grasp matrix of thumb ${}^o\bar{\mathbf{G}} \in \Re^{6\times \bar{n}\bar{c}}$ (nc and $\bar{n}\bar{c}$ represent the total number of force factors, including number of force vectors and type of forces at time t), described as

$$\mathbf{w}^o(t) + {}^o\mathbf{G}\mathbf{f}^o_{ht} + {}^o\bar{\mathbf{G}}\bar{\mathbf{f}}^o_{ht} = \mathbf{0}. \tag{13}$$

Calculating the differential of (13), perturbed equilibrium equation of the object can be obtained as

$$\delta\mathbf{w}^o(t) + {}^o\mathbf{G}\delta\mathbf{f}^o_{ht} + {}^o\bar{\mathbf{G}}\delta\bar{\mathbf{f}}^o_{ht} = \mathbf{0}. \tag{14}$$

Since the contact force variations $\delta\mathbf{f}^o_{ht}$ and $\delta\bar{\mathbf{f}}^o_{ht}$ are determined by two independent actuating modules, in the followings, we introduce quasi-static equations describing the fingers-object system and thumb-object system, respectively.

Fingers-object system: By utilizing a linear contact force model in the interface [36], the contact force variation $\delta\mathbf{f}^o_{ht}$ can be related to the contact position variations through a stiffness matrix $\mathbf{K}_C \in \Re^{nc\times nc}$, which describes the characteristics of virtual springs. The contact positions can be represented by

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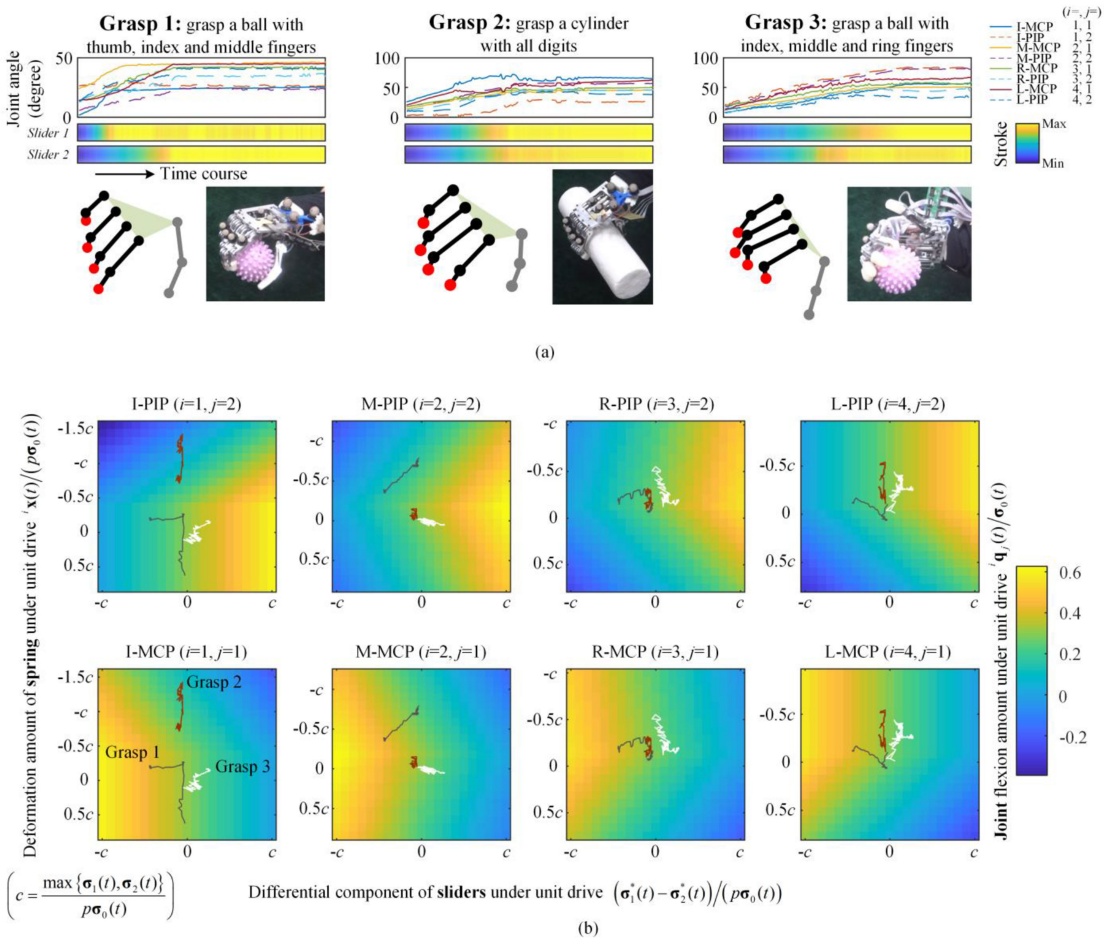


Fig. 7. Visualization of self-adaptively grasping processes of the underactuated differential mechanism with three examples. (a) Experimental examples: three typical self-adaptive grasps and their joint angle data, slider stroke data, simulation postures, and practical grasping postures. I-MCP refers to the MCP joint of index finger. M: middle finger; R: ring finger; L: little finger. (b) Visualization of the self-adaptively grasping processes. The eight figures visualize (12), which puts the rotation angle of the motor (active element) $\sigma_0(t)$ in denom and highlights the eight joint angle variations with respect to variations of displacements of two passive elements: 1) the differential component of sliders (displacement difference between the two sliders) and 2) the deformation amounts of springs. All numerical values are then dimensionless. The constant $c = 1$ appears in general situations, $c \neq 1$ only appears in the situation that the motor moves but both sliders stop moving (usually the end of self-adaptive grasping). The backgrounds of figures show all possible situations and the three curves named Grasp 1–3 correspond to the three examples in (a). Dynamic variations of the curves with time during grasping processes can be found in the supplementary video. Note that, the moving directions of the curves in the plane-coordinate system constructed by displacements of passive elements are able to reflect self-adaptive movements of the underactuated differential mechanism in the mini X-hand.

both $\mathbf{C}_{ht} \in \Re^{nc}$, which is determined by configuration of four fingers $\mathbf{Q}(t) \in \Re^8$, and $\mathbf{C}_{ot} \in \Re^{nc}$, which is determined by object configuration $\mathbf{u}(t) \in \Re^6$. The relationship is as follows:

$$\delta \mathbf{f}_{ht}^o = \mathbf{K}_C \left(\delta \mathbf{C}_{ht} - \delta \mathbf{C}_{ot} \right) \tag{15}$$

with

$$\delta \mathbf{C}_{ot} = {}^o\mathbf{G}^T \delta \mathbf{u}(t) \tag{16}$$

$$\delta \mathbf{C}_{ht} = \mathbf{J}\left(\mathbf{Q}(t), \mathbf{u}(t)\right) \delta \mathbf{Q}(t). \tag{17}$$

Here, $\mathbf{J}(\mathbf{Q}(t), \mathbf{u}(t)) \in \Re^{nc \times 8}$ is hand Jacobian matrix in object framework, which also maps contact forces $\mathbf{f}_{ht}^o$ to joint torque $\tau(t) \in \Re^{8\times 1}$ in the following term:

$$\tau(t) = \mathbf{J}^T \left(\mathbf{Q}(t), \mathbf{u}(t)\right) \mathbf{f}_{ht}^o. \tag{18}$$

Calculating the differential of (18), the joint torque variation $\delta\tau(t)$ is determined by the contact force variation $\delta \mathbf{f}_{ht}^o$, object configuration variation $\delta \mathbf{u}(t)$, and joint angle variation $\delta \mathbf{Q}(t)$, as follows:

$$\delta\tau(t) = \mathbf{\Omega}_{\mathbf{Q}} \delta \mathbf{Q}(t) + \mathbf{\Omega}_{\mathbf{u}} \delta \mathbf{u}(t) + \mathbf{J}^T \left(\mathbf{Q}(t), \mathbf{u}(t)\right) \delta \mathbf{f}_{ht}^o \tag{19}$$

where

$$\mathbf{\Omega}_{\mathbf{Q}} = \frac{\partial \mathbf{J}^T \left(\mathbf{Q}(t), \mathbf{u}(t)\right) \mathbf{f}_{ht}^o}{\partial \mathbf{Q}(t)} \text{ and } \mathbf{\Omega}_{\mathbf{u}} = \frac{\partial \mathbf{J}^T \left(\mathbf{Q}(t), \mathbf{u}(t)\right) \mathbf{f}_{ht}^o}{\partial \mathbf{u}(t)}. \tag{20}$$

The joint angle variation $\delta \mathbf{Q}(t)$ is further determined by the variations of displacements of sliders $\delta \sigma_j^*(t)$ $(j = 1, 2)$ and passive deformations of springs $\delta \mathbf{x}(t)$ with $\mathbf{x} = \left[{}^1\mathbf{x}^T \ {}^2\mathbf{x}^T \ {}^3\mathbf{x}^T \ {}^4\mathbf{x}^T \right]^T \in \Re^{4\times T}$. Calculating the differential of (10), and letting ${}^i\mathbf{M}_j \in \Re^{2\times 1}$ $(j = 1, 2)$ be the jth row of ${}^i\mathbf{M}$, we have

$$\delta \mathbf{Q}(t) = \mathbf{Tr}_{\sigma *} \begin{bmatrix} \delta \sigma_1^*(t) \\ \delta \sigma_2^*(t) \end{bmatrix} + \mathbf{Tr}_{\mathbf{x}} \delta \mathbf{x}(t) \tag{21}$$

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TABLE I
NOTATIONS FOR THE DUAL-ACTUATED GRASP AND MANIPULATION ANALYSIS

Notation	Definition
$\delta\mathbf{y}$	Variation of variable $\mathbf{y}$
$\bar{\mathbf{y}}$	Variable in thumb-object system
$\mathbf{w}^o(t) \in \Re^6$	External wrench acting on the object
$\mathbf{u}(t) \in \Re^6$	Object configuration
$\mathbf{C}_{ot} \in \Re^{nc}$, $\bar{\mathbf{C}}_{ot} \in \Re^{\overline{nc}}$	Contact positions determined by object
$\mathbf{C}_{ht} \in \Re^{nc}$, $\bar{\mathbf{C}}_{ht} \in \Re^{\overline{nc}}$	Contact positions determined by hand
$\mathbf{f}_{ht}^o \in \Re^{nc}$, $\bar{\mathbf{f}}_{ht}^o \in \Re^{\overline{nc}}$	Contact forces by hand on the object
nc , $\overline{nc}$	Numbers of contact constraints
$\boldsymbol{\tau}(t) \in \Re^8$, $\bar{\boldsymbol{\tau}}(t) \in \Re^3$	Joint torques
$\mathbf{x}(t) \in \Re^4$, $\bar{\mathbf{x}}(t) \in \Re$	Displacements of springs
$\mathbf{Q}(t) \in \Re^8$, $\bar{\mathbf{Q}}(t) \in \Re^3$	Joint angles
$\mathbf{F}_j^{slider}(t) \in \Re$	Actuating force of the j th slider
$\mathbf{c}_j(t) \in \Re$	Relative rotation angle at both ends of the jth drag axis
$\boldsymbol{\sigma}_j^*(t) \in \Re$	Displacement of the j th slider
$\boldsymbol{\sigma}_0(t) \in \Re$, $\bar{\boldsymbol{\sigma}}_0(t) \in \Re$	Rotation angles of motor shafts
$\boldsymbol{\tau}^{input}(t) \in \Re$, $\bar{\boldsymbol{\tau}}^{input}(t) \in \Re$	Output torques of motors
${}^o\mathbf{G} \in \Re^{6\times nc}$, ${}^o\bar{\mathbf{G}} \in \Re^{6\times\overline{nc}}$	Grasp matrices
$\mathbf{J}(\mathbf{Q}(t),\ \mathbf{u}(t)) \in \Re^{nc\times 8}$, $\bar{\mathbf{J}}(\bar{\mathbf{Q}}(t),\ \mathbf{u}(t)) \in \Re^{\overline{nc}\times 3}$	Jacobian matrices in object framework
$\boldsymbol{\Omega}_{\mathbf{y}}$, $\bar{\boldsymbol{\Omega}}_{\mathbf{y}}$, $\bar{\boldsymbol{\Omega}}_{\bar{\mathbf{y}}}$	Partial differential items $\partial\boldsymbol{\tau}/\partial\mathbf{y}$, $\partial\bar{\boldsymbol{\tau}}/\partial\mathbf{y}$ and $\partial\bar{\boldsymbol{\tau}}/\partial\bar{\mathbf{y}}$
$\mathbf{Tr}_{\mathbf{y}}$	Motion transmission matrix with motion input $\mathbf{y}$
$\mathbf{K}_C \in \Re^{nc\times nc}$, $\bar{\mathbf{K}}_C \in \Re^{\overline{nc}\times\overline{nc}}$	Stiffness at contact positions
$\mathbf{K} \in \Re^{4\times 4}$, $\bar{K}$	Stiffness of springs

where

$$\mathbf{Tr}_{\sigma *} = \text{blkdiag}\left\{{}^1\mathbf{M}_1, {}^2\mathbf{M}_1, {}^3\mathbf{M}_1, {}^4\mathbf{M}_1\right\}\mathbf{N} \in \Re^{8\times 2}$$

$$\mathbf{Tr}_{\mathbf{x}} = \text{blkdiag}\left\{{}^1\mathbf{M}_2, {}^2\mathbf{M}_2, {}^3\mathbf{M}_2, {}^4\mathbf{M}_2\right\} \in \Re^{8\times 4}. \quad (22)$$

Correspondingly, force variation relationship among joints, sliders, and springs can be obtained based on the dual principle

$$\begin{bmatrix}\delta\mathbf{F}_1^{\text{slider}}(t)\\ \delta\mathbf{F}_2^{\text{slider}}(t)\end{bmatrix} = \mathbf{Tr}_{\sigma *}{}^T\delta\tau(t) \quad (23)$$

$$\mathbf{K}\delta\mathbf{x}(t) = \mathbf{Tr}_{\mathbf{x}}{}^T\delta\tau(t) \Rightarrow \delta\mathbf{x}(t) = \mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\delta\tau(t) \quad (24)$$

where $\mathbf{K} = \text{diag}\{{}^1k, {}^2k, {}^3k, {}^4k\} \in \Re^{4\times 4}$ indicates the stiffness matrix of springs.

In the link of motor and sliders, given the variation of rotation angle of the motor shaft (motion input variation) $\delta\sigma_0(t)$, according to the differential of (8), variations of displacements of sliders $\delta\sigma_j^*(t)$ $(j = 1, 2)$ are determined as

$$\begin{bmatrix}\delta\sigma_1^*(t)\\ \delta\sigma_2^*(t)\end{bmatrix} = \mathbf{Tr}_{\sigma_0}\delta\sigma_0(t) + \mathbf{Tr}_{\mathbf{c}}\begin{bmatrix}\delta\mathbf{c}_1(t)\\ \delta\mathbf{c}_2(t)\end{bmatrix} \quad (25)$$

with

$$\mathbf{Tr}_{\sigma_0} = \begin{bmatrix}p\\ p\end{bmatrix} \in \Re^{2\times 1}, \mathbf{Tr}_{\mathbf{c}} = \begin{bmatrix}-p & \\ & -p\end{bmatrix} \in \Re^{2\times 2}. \quad (26)$$

Here, $\delta\mathbf{c}_j(t)$ $(j = 1, 2)$ are variations of relative rotation angles at both ends of the jth drag axis, i.e., passive rotations of drag axes, determined by the forces exerting on the drag axes $\mathbf{F}_j^{\text{slider}}(t)$ and rotation direction of the motor shift

$$\delta\mathbf{c}_j(t) = \Gamma_j\left(\mathbf{F}_j^{\text{slider}}(t), \delta\sigma_0(t)\right)\delta\sigma_0(t) \quad (27)$$

where

$$\Gamma_j\left(\mathbf{F}_j^{\text{slider}}(t), \delta\sigma_0(t)\right) = \begin{cases}1, & \mathbf{F}_j^{\text{slider}}(t) = \tau_{\max}^{\text{dragaxis}}/p \\ & \text{and } \delta\sigma_0(t) > 0 \\ 0, & \text{else}\end{cases}. \quad (28)$$

Differentiating (9) and introducing the symbol $\mathbf{Tr}_{\sigma_0}$ of (26), the relationship between motor torque variation and actuating force variations of sliders is obtained as follows:

$$\delta\tau^{\text{input}}(t) = \mathbf{Tr}_{\sigma_0}{}^T\begin{bmatrix}\delta\mathbf{F}_1^{\text{slider}}(t)\\ \delta\mathbf{F}_2^{\text{slider}}(t)\end{bmatrix}. \quad (29)$$

Quasi-static equations in each link of the fingers-object system have been carried out.

Thumb-object system: Similar to (15), contact force variations of thumb $\delta\bar{\mathbf{f}}_{ht}^o$ can be represented by contact position variations as follows:

$$\delta\bar{\mathbf{f}}_{ht}^o = \bar{\mathbf{K}}_C\left(\delta\bar{\mathbf{C}}_{ht} - \delta\bar{\mathbf{C}}_{ot}\right) \quad (30)$$

with

$$\delta\bar{\mathbf{C}}_{ot} = {}^o\bar{\mathbf{G}}^T\delta\mathbf{u}(t) \quad (31)$$

$$\delta\bar{\mathbf{C}}_{ht} = \bar{\mathbf{J}}\left(\bar{\mathbf{Q}}(t), \mathbf{u}(t)\right)\delta\bar{\mathbf{Q}}(t). \quad (32)$$

The thumb configuration variation $\delta\bar{\mathbf{Q}}(t)$ is further determined by the variation of rotation angle of the motor shaft $\delta\bar{\sigma}_0(t)$ and variation of passive deformations of springs $\delta\bar{\mathbf{x}}(t)$

$$\delta\bar{\mathbf{Q}}(t) = \mathbf{Tr}_{\bar{\sigma}_0}\delta\bar{\sigma}_0(t) + \mathbf{Tr}_{\bar{\mathbf{x}}}\delta\bar{\mathbf{x}}(t). \quad (33)$$

Variation of passive deformations of springs $\delta\bar{\mathbf{x}}(t)$ is further determined by

$$\delta\bar{\mathbf{x}}(t) = \bar{K}^{-1}\mathbf{Tr}_{\bar{\mathbf{x}}}{}^T\delta\bar{\tau}(t) \quad (34)$$

where

$$\delta\bar{\tau}(t) = \bar{\boldsymbol{\Omega}}_{\bar{\mathbf{Q}}}\delta\bar{\mathbf{Q}}(t) + \bar{\boldsymbol{\Omega}}_{\mathbf{u}}\delta\mathbf{u}(t) + \bar{\mathbf{J}}^T\left(\bar{\mathbf{Q}}(t), \mathbf{u}(t)\right)\delta\bar{\mathbf{f}}_{ht}^o. \quad (35)$$

In addition, joint torque variation $\delta\bar{\tau}(t)$ further determines motor torque variation in the following term:

$$\delta\bar{\tau}^{input}(t) = \mathbf{Tr}_{\bar{\sigma}_0}{}^T\delta\bar{\tau}(t). \quad (36)$$

Quasi-static equations in all links of the hand-object system have been carried out. Theses quasi-static equations are visually presented in Fig. 5. This figure can help readers to intuitively understand the movement and force relationships in the hand-object system on the one hand, and is convenient to deduce relationships between critical variables on the other hand (as follows).

B. Free Movements of the Robotic Hand

Before the hand contacting the object, the numbers of contacts $nc = \bar{n}\bar{c} = 0$, which means that $\mathbf{C}_{ht}, \mathbf{C}_{ot}, \bar{\mathbf{C}}_{ht}, \bar{\mathbf{C}}_{ot}$ do not exist, so that all passive elements do not work. According to Fig. 5, the

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configuration variations reflecting free movements of the hand $\delta\mathbf{Q}_{\text{free}}(t) \in \Re^8$ and $\delta\bar{\mathbf{Q}}_{\text{free}}(t) \in \Re^3$ can be described as follows:

$$\begin{cases} \delta\mathbf{Q}_{\text{free}}(t) = \mathbf{Tr}_{\sigma*}\mathbf{Tr}_{\sigma_0}\delta\sigma_0(t) \\ \delta\bar{\mathbf{Q}}_{\text{free}}(t) = \mathbf{Tr}_{\bar{\sigma}_0}\delta\bar{\sigma}_0(t) \end{cases}. \tag{37}$$

C. Induction of Passive Movements by Contact Forces

After contacting the object, the contact forces would induce movements of some passive elements. At this time, according to Fig. 5 [combining (21) and (25)], the motions of fingers can be integrally described as follows:

$$\delta\mathbf{Q}(t) = \underbrace{\mathbf{Tr}_{\sigma*}\mathbf{Tr}_{\sigma_0}\delta\sigma_0(t)}_{\delta\mathbf{Q}_{\text{free}}(t)} + \underbrace{\mathbf{Tr}_{\mathbf{x}}\delta\mathbf{x}(t) + \mathbf{Tr}_{\sigma*}\mathbf{Tr}_{\mathbf{c}}\begin{bmatrix}\delta\mathbf{c}_1(t)\\ \delta\mathbf{c}_2(t)\end{bmatrix}}_{\delta\mathbf{Q}_{\text{passive}}(t)} \tag{38}$$

where configuration variation $\delta\mathbf{Q}_{\text{passive}}(t) \in \Re^8$ reflects passive movements of fingers, decided by the deformations of springs and relative rotations of two ends in drag axes. Combining (27), (28), (18), and (23), the configuration variations of drag axes can be described as follows:

$$\begin{bmatrix}\delta\mathbf{c}_1(t)\\ \delta\mathbf{c}_2(t)\end{bmatrix} = \begin{bmatrix}\Gamma_1\left(\mathbf{F}_1^{\text{slider}}(t), \delta\sigma_0(t)\right)\\ \Gamma_2\left(\mathbf{F}_2^{\text{slider}}(t), \delta\sigma_0(t)\right)\end{bmatrix}\delta\sigma_0(t) \tag{39}$$

where

$$\Gamma_j\left(\mathbf{F}_j^{\text{slider}}(t), \delta\sigma_0(t)\right) = \begin{cases} 1, & \mathbf{F}_j^{\text{slider}}(t) = \mathbf{Tr}_{\sigma*j}{}^T\mathbf{J}^T\left(\mathbf{Q}(t), \mathbf{u}(t)\right)\mathbf{f}_{ht}^o = \tau_{\max}^{\text{dragaxis}}/p \\ & \text{and } \delta\sigma_0(t) > 0 \\ 0, & \text{else} \end{cases}. \tag{40}$$

Note that $\mathbf{Tr}_{\sigma*j} \in \Re^{8\times 1}$ are the jth column of $\mathbf{Tr}_{\sigma*}$ $(j = 1, 2)$.

Combining (24) and (19), the passive deformations of springs can be described as

$$\begin{aligned}&\delta\mathbf{x}(t) = \mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\cdot \\ &\left(\mathbf{\Omega}_{\mathbf{Q}}\delta\mathbf{Q}(t) + \mathbf{\Omega}_{\mathbf{u}}\delta\mathbf{u}(t) + \mathbf{J}^T\left(\mathbf{Q}(t), \mathbf{u}(t)\right)\delta\mathbf{f}_{ht}^o\right).\end{aligned} \tag{41}$$

To eliminate $\delta\mathbf{Q}(t)$, which is conversely determined by $\delta\mathbf{x}(t)$, we substitute (38) and (39) into (41), and obtain

$$\delta\mathbf{x}(t) = \mathbf{tr}_{\sigma_0}^{\mathbf{x}}\delta\sigma_0(t) + \mathbf{tr}_{\mathbf{u}}^{\mathbf{x}}\delta\mathbf{u}(t) + \mathbf{tr}_{\mathbf{f}_h^o}^{\mathbf{x}}\delta\mathbf{f}_{ht}^o \tag{42}$$

where

$$\begin{aligned}\mathbf{tr}_{\sigma_0}^{\mathbf{x}} &= \left(\mathbf{I} - \mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\mathbf{\Omega}_{\mathbf{Q}}\mathbf{Tr}_{\mathbf{x}}\right)^{-1}\mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\mathbf{\Omega}_{\mathbf{Q}}\cdot \\ &\quad \mathbf{Tr}_{\sigma*}\left(\mathbf{Tr}_{\sigma_0} + \mathbf{Tr}_{\mathbf{c}}\begin{bmatrix}\Gamma_1\left(\mathbf{F}_1^{\text{slider}}(t), \delta\sigma_0(t)\right)\\ \Gamma_2\left(\mathbf{F}_2^{\text{slider}}(t), \delta\sigma_0(t)\right)\end{bmatrix}\right) \in \Re^{4\times 1} \\ \mathbf{tr}_{\mathbf{u}}^{\mathbf{x}} &= \left(\mathbf{I} - \mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\mathbf{\Omega}_{\mathbf{Q}}\mathbf{Tr}_{\mathbf{x}}\right)^{-1}\mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\mathbf{\Omega}_{\mathbf{u}} \in \Re^{4\times 6} \\ \mathbf{tr}_{\mathbf{f}_h^o}^{\mathbf{x}} &= \left(\mathbf{I} - \mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\mathbf{\Omega}_{\mathbf{Q}}\mathbf{Tr}_{\mathbf{x}}\right)^{-1}\mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\mathbf{J}^T\left(\mathbf{Q}(t), \mathbf{u}(t)\right) \\ &\quad \in \Re^{4\times nc}.\end{aligned} \tag{43}$$

Substituting (39) and (42) into (38), the passive movements of fingers represented by configuration variation $\delta\mathbf{Q}_{\text{passive}}(t)$ can be finally described as

$$\begin{aligned}\delta\mathbf{Q}_{\text{passive}}(t) &= \mathbf{Tr}_{\mathbf{x}}\left(\mathbf{tr}_{\sigma_0}^{\mathbf{x}}\delta\sigma_0(t) + \mathbf{tr}_{\mathbf{u}}^{\mathbf{x}}\delta\mathbf{u}(t) + \mathbf{tr}_{\mathbf{f}_h^o}^{\mathbf{x}}\delta\mathbf{f}_{ht}^o\right) \\ &\quad + \mathbf{Tr}_{\sigma*}\mathbf{Tr}_{\mathbf{c}}\begin{bmatrix}\Gamma_1\left(\mathbf{F}_1^{\text{slider}}(t), \delta\sigma_0(t)\right)\\ \Gamma_2\left(\mathbf{F}_2^{\text{slider}}(t), \delta\sigma_0(t)\right)\end{bmatrix}\delta\sigma_0(t).\end{aligned} \tag{44}$$

For thumb, combining (33), (34), and (35), the passive deformation of the spring is obtained as

$$\delta\bar{\mathbf{x}}(t) = tr_{\bar{\sigma}_0}^{\bar{\mathbf{x}}}\delta\bar{\sigma}_0(t) + \mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{x}}}\delta\mathbf{u}(t) + \mathbf{tr}_{\bar{\mathbf{f}}_h^o}^{\bar{\mathbf{x}}}\delta\bar{\mathbf{f}}_{ht}^o \tag{45}$$

with

$$\begin{aligned}tr_{\bar{\sigma}_0}^{\bar{\mathbf{x}}} &= \left(1 - \bar{K}^{-1}\mathbf{Tr}_{\bar{\mathbf{x}}}{}^T\bar{\mathbf{\Omega}}_{\bar{\mathbf{Q}}}\mathbf{Tr}_{\bar{\mathbf{x}}}\right)^{-1}\bar{K}^{-1}\mathbf{Tr}_{\bar{\mathbf{x}}}{}^T\bar{\mathbf{\Omega}}_{\bar{\mathbf{Q}}}\mathbf{Tr}_{\bar{\sigma}_0} \in \Re \\ \mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{x}}} &= \left(1 - \bar{K}^{-1}\mathbf{Tr}_{\bar{\mathbf{x}}}{}^T\bar{\mathbf{\Omega}}_{\bar{\mathbf{Q}}}\mathbf{Tr}_{\bar{\mathbf{x}}}\right)^{-1}\bar{K}^{-1}\mathbf{Tr}_{\bar{\mathbf{x}}}{}^T\bar{\mathbf{\Omega}}_{\mathbf{u}} \in \Re^{1\times 6} \\ \mathbf{tr}_{\bar{\mathbf{f}}_h^o}^{\bar{\mathbf{x}}} &= \left(1 - \bar{K}^{-1}\mathbf{Tr}_{\bar{\mathbf{x}}}{}^T\bar{\mathbf{\Omega}}_{\bar{\mathbf{Q}}}\mathbf{Tr}_{\bar{\mathbf{x}}}\right)^{-1}\bar{K}^{-1}\mathbf{Tr}_{\bar{\mathbf{x}}}{}^T\bar{\mathbf{J}}^T\left(\bar{\mathbf{Q}}(t),\right. \\ &\quad \left.\mathbf{u}(t)\right) \in \Re^{1\times\bar{n}\bar{c}}.\end{aligned} \tag{46}$$

Substituting (45) into (33), and defining configuration variation $\delta\bar{\mathbf{Q}}_{\text{passive}}(t) = \mathbf{Tr}_{\bar{\mathbf{x}}}\delta\bar{\mathbf{x}}(t)$, we have

$$\delta\bar{\mathbf{Q}}_{\text{passive}}(t) = \mathbf{Tr}_{\bar{\mathbf{x}}}\left(tr_{\bar{\sigma}_0}^{\bar{\mathbf{x}}}\delta\bar{\sigma}_0(t) + \mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{x}}}\delta\mathbf{u}(t) + \mathbf{tr}_{\bar{\mathbf{f}}_h^o}^{\bar{\mathbf{x}}}\delta\bar{\mathbf{f}}_{ht}^o\right). \tag{47}$$

Combining (42) and (47), the relationships between configuration variations reflecting passive movements of the robotic hand and contact force variations can be integrated in (55). It is clear that the passive movements of the hand are not only induced by the contact force variations but also determined by the motion input variations of the motors and object movements. In addition, the transmission matrix is also related to the actuating forces of the sliders $\mathbf{F}_j^{\text{slider}}(t)$, hand configuration $\mathbf{Q}(t)$ [related to $\mathbf{tr}_{\mathbf{f}_h^o}^{\mathbf{x}}$ according to (43)], $\bar{\mathbf{Q}}(t)$ [related to $\mathbf{tr}_{\bar{\mathbf{f}}_h^o}^{\bar{\mathbf{x}}}$ according to (46)] and object configuration $\mathbf{u}(t)$ (related to both $\mathbf{tr}_{\mathbf{f}_h^o}^{\mathbf{x}}$ and $\mathbf{tr}_{\bar{\mathbf{f}}_h^o}^{\bar{\mathbf{x}}}$), which are also the determination factors of the passive movements at time t.

D. Dual-Actuated Grasping and Manipulation

Combining (15), (16), and (17), we can reach

$$\delta\mathbf{f}_{ht}^o = \mathbf{K}_C\mathbf{J}\left(\mathbf{Q}(t), \mathbf{u}(t)\right)\delta\mathbf{Q}(t) - \mathbf{K}_C{}^o\mathbf{G}^T\delta\mathbf{u}(t). \tag{48}$$

Substituting (41) and (39) into (38) and substituting (38) into (48), $\delta\mathbf{Q}(t)$ can be eliminated, and the contact force variation $\delta\mathbf{f}_{ht}^o$ is determined by motion input variation $\delta\sigma_0(t)$ and object configuration variation $\delta\mathbf{u}(t)$ as follows:

$$\delta\mathbf{f}_{ht}^o = \mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o}\delta\sigma_0(t) + \mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o}\delta\mathbf{u}(t) \tag{49}$$

with

$$\begin{aligned}\mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} &= \begin{pmatrix}\mathbf{I} - \mathbf{K}_C\mathbf{J}\left(\mathbf{Q}(t), \mathbf{u}(t)\right)\left(\mathbf{I} - \mathbf{Tr}_{\mathbf{x}}\mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\mathbf{\Omega}_{\mathbf{Q}}\right)^{-1}\\ \mathbf{Tr}_{\mathbf{x}}\mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\mathbf{J}^T\left(\mathbf{Q}(t), \mathbf{u}(t)\right)\end{pmatrix}^{-1} \\ &\quad \cdot \mathbf{K}_C\mathbf{J}\left(\mathbf{Q}(t), \mathbf{u}(t)\right)\left(\mathbf{I} - \mathbf{Tr}_{\mathbf{x}}\mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\mathbf{\Omega}_{\mathbf{Q}}\right)^{-1}\mathbf{Tr}_{\sigma*} \\ &\quad \cdot \left(\mathbf{Tr}_{\sigma_0} + \mathbf{Tr}_{\mathbf{c}}\begin{bmatrix}\Gamma_1\left(\mathbf{F}_1^{\text{slider}}(t), \delta\sigma_0(t)\right)\\ \Gamma_2\left(\mathbf{F}_2^{\text{slider}}(t), \delta\sigma_0(t)\right)\end{bmatrix}\right) \in \Re^{nc\times 1} \\ \mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} &= \begin{pmatrix}\mathbf{I} - \mathbf{K}_C\mathbf{J}\left(\mathbf{Q}(t), \mathbf{u}(t)\right)\left(\mathbf{I} - \mathbf{Tr}_{\mathbf{x}}\mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\mathbf{\Omega}_{\mathbf{Q}}\right)^{-1}\\ \mathbf{Tr}_{\mathbf{x}}\mathbf{K}^{-1}\mathbf{Tr}_{\mathbf{x}}{}^T\mathbf{J}^T\left(\mathbf{Q}(t), \mathbf{u}(t)\right)\end{pmatrix}^{-1}\end{aligned}$$

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$$\cdot \begin{pmatrix} \mathbf{K}_C \mathbf{J}\left(\mathbf{Q}(t), \mathbf{u}(t)\right)\left(\mathbf{I} - \mathbf{Tr}_{\mathbf{x}} \mathbf{K}^{-1} \mathbf{Tr}_{\mathbf{x}}{}^T \mathbf{\Omega}_{\mathbf{Q}}\right)^{-1} \\ \mathbf{Tr}_{\mathbf{x}} \mathbf{K}^{-1} \mathbf{Tr}_{\mathbf{x}}{}^T \mathbf{\Omega}_{\mathbf{u}} - \mathbf{K}_C {}^o\mathbf{G}^T \end{pmatrix} \in \Re^{nc \times 6}. \quad (50)$$

Equation (49) indicates that if the object keeps immobile (configure variation $\delta \mathbf{u} = \mathbf{0}$), it is possible to adjust the contact forces $\mathbf{f}_{ht}^o$ through changing the motion input $\sigma_0(t)$ under the condition $\mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} \neq \mathbf{0}$. Easily we can reach that the contact forces $\mathbf{f}_{ht}^o$ relate to the actuating forces of sliders as $\begin{bmatrix} \mathbf{F}_1^{\text{slider}}(t) \\ \mathbf{F}_2^{\text{slider}}(t) \end{bmatrix} = \mathbf{Tr}_{\sigma *}{}^T \mathbf{J}^T(\mathbf{Q}(t), \mathbf{u}(t)) \mathbf{f}_{ht}^o$ according to (18) and (23), which means that the actuating forces of sliders are related to the motion input. With the motion input increased, the actuating forces of the two sliders would no doubt increase. If at time t, the two drag axes both reach the slipping torque $\tau_{\max}^{\text{dragaxis}}$, that is $\Gamma_1 = \Gamma_2 = 1$, then $\mathbf{Tr}_{\sigma_0} + \mathbf{Tr}_{\mathbf{c}} \begin{bmatrix} \Gamma_1 \\ \Gamma_2 \end{bmatrix} = \begin{bmatrix} p \\ p \end{bmatrix} + \begin{bmatrix} -p \\ -p \end{bmatrix} = \mathbf{0}$ [according to (26)], which means that $\mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} = \mathbf{0}$ [according to (50)]. From this moment on, if the motion input variable keeps increased ($\delta\sigma_0 > \mathbf{0}$), $\mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o}$ would keep zero. The contact forces would not change any more no matter how to increase the current or rotation angle of the motor. In other words, the robotic hand has the grasping limitations that are decided by the slipping torque $\tau_{\max}^{\text{dragaxis}}$ of drag axes. Defining the slipping torque $\tau_{\max}^{\text{dragaxis}}$ as large as possible is desirable, but it is limited by the maximum motor torque: if $\tau_{\max}^{\text{dragaxis}}$ is too large, the maximum torque of the motor would be lower than $\mathbf{Tr}_{\sigma_0}{}^T \begin{bmatrix} \tau_{\max}^{\text{dragaxis}}/p \\ \tau_{\max}^{\text{dragaxis}}/p \end{bmatrix}$ so that (9) and (29) cannot be satisfied. One reasonable scheme may be: 1) choosing as much torque as possible for motors under size constraints, and 2) choosing as large slipping torque as possible for the drag axes under the constraint of the chosen maximum motor torque.

Let us shift our focus from fingers to thumb. Combining (30), (31), and (32), we can reach

$$\delta \bar{\mathbf{f}}_{ht}^o = \bar{\mathbf{K}}_C \bar{\mathbf{J}}\left(\bar{\mathbf{Q}}(t), \mathbf{u}(t)\right) \delta \bar{\mathbf{Q}}(t) - \bar{\mathbf{K}}_C {}^o\bar{\mathbf{G}}^T \delta \mathbf{u}(t). \quad (51)$$

Substituting (35) into (34), then substituting (34) into (33), and finally substituting (33) into (51), we have

$$\delta \bar{\mathbf{f}}_{ht}^o = \mathbf{tr}_{\bar{\sigma}_0}^{\bar{\mathbf{f}}_h^o} \delta \bar{\sigma}_0(t) + \mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o} \delta \mathbf{u}(t) \quad (52)$$

where

$$\mathbf{tr}_{\bar{\sigma}_0}^{\bar{\mathbf{f}}_h^o} = \left(\mathbf{I} - \bar{\mathbf{K}}_C \bar{\mathbf{J}}\left(\bar{\mathbf{Q}}, \mathbf{u}\right)\left(1 - \bar{K}^{-1} \mathbf{Tr}_{\bar{\mathbf{x}}}{}^T \bar{\mathbf{\Omega}}_{\bar{\mathbf{Q}}} \mathbf{Tr}_{\bar{\mathbf{x}}}\right)^{-1} \mathbf{Tr}_{\bar{\mathbf{x}}} \mathbf{Tr}_{\bar{\mathbf{f}}_h^o}^{\bar{\mathbf{x}}}\right)^{-1} \cdot \bar{\mathbf{K}}_C \bar{\mathbf{J}}\left(\bar{\mathbf{Q}}, \mathbf{u}\right)\left(1 - \bar{K}^{-1} \mathbf{Tr}_{\bar{\mathbf{x}}}{}^T \bar{\mathbf{\Omega}}_{\bar{\mathbf{Q}}} \mathbf{Tr}_{\bar{\mathbf{x}}}\right)^{-1} \times \left(\mathbf{Tr}_{\bar{\mathbf{x}}} \mathbf{Tr}_{\bar{\sigma}_0}^{\bar{\mathbf{x}}} + \mathbf{Tr}_{\bar{\sigma}_0}\right) \in \Re^{\bar{n}\bar{c} \times 1}$$

$$\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o} = \left(\mathbf{I} - \bar{\mathbf{K}}_C \bar{\mathbf{J}}\left(\bar{\mathbf{Q}}, \mathbf{u}\right)\left(1 - \bar{K}^{-1} \mathbf{Tr}_{\bar{\mathbf{x}}}{}^T \bar{\mathbf{\Omega}}_{\bar{\mathbf{Q}}} \mathbf{Tr}_{\bar{\mathbf{x}}}\right)^{-1} \mathbf{Tr}_{\bar{\mathbf{x}}} \mathbf{Tr}_{\bar{\mathbf{f}}_h^o}^{\bar{\mathbf{x}}}\right)^{-1} \cdot \left(-\bar{\mathbf{K}}_C {}^o\bar{\mathbf{G}}^T + \bar{\mathbf{K}}_C \bar{\mathbf{J}}\left(\bar{\mathbf{Q}}, \mathbf{u}\right)\left(1 - \bar{K}^{-1} \mathbf{Tr}_{\bar{\mathbf{x}}}{}^T \bar{\mathbf{\Omega}}_{\bar{\mathbf{Q}}} \mathbf{Tr}_{\bar{\mathbf{x}}}\right)^{-1} \times \mathbf{Tr}_{\bar{\mathbf{x}}} \mathbf{Tr}_{\mathbf{u}}^{\bar{\mathbf{x}}}\right) \in \Re^{\bar{n}\bar{c} \times 6}. \quad (53)$$

TABLE II
MECHANISM PARAMETERS OF THE MINI X-HAND (UNIT: mm OR N/mm)

i =	${}^i r_1$	${}^i r_2$	${}^i r_1'$	${}^i r_2'$	${}^i k$	d_{1i}
1(index)	4.1	4.1	6.4	3.3	0.8	0
2(middle)	3.9	3.9	4.7	3.3	0.8	18
3(ring)	4.4	4.4	4.4	3.3	0.8	37.7
4(little)	4.3	4.3	4	3.3	0.8	47
5(thumb)	4.6	3.9	3.4	3.9	0.8	/

Substituting (49) and (52) into (14), it can be found that the object configuration variation $\delta \mathbf{u}(t)$ is determined by the two motion input variations $\delta\sigma_0(t)$ and $\delta\bar{\sigma}_0(t)$ as well as external wrench variation $\delta \mathbf{w}^o$, described as follows:

$$\delta \mathbf{u}(t) = -\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} \times \left(\delta \mathbf{w}^o + {}^o\mathbf{G}\mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} \delta\sigma_0(t) + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\bar{\sigma}_0}^{\bar{\mathbf{f}}_h^o} \delta\bar{\sigma}_0(t)\right). \quad (54)$$

Substituting (54) into (49) and (52), we can reach (56) shown at the bottom of the next page. Equation (56) indicates that if not involving external wrench variations, the contact force variations and object configuration variations can be implemented by the two motion input variations, which means that it is possible to adjust grasping forces and to implement in-hand manipulation through changing the rotation angles of two motor shifts. At this time, the transmission matrix in (56) reflects the grasping and manipulation properties. The span of matrix, unnumbered equation shown at the bottom of this page, is the subspace of controllable internal force, and the range space of

$$\left[-\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\mathbf{G}\mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} - \left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\bar{\mathbf{G}}\mathbf{tr}_{\bar{\sigma}_0}^{\bar{\mathbf{f}}_h^o}\right]$$

represents the subspace of the controllable internal object displacements according to the literature [36]. In the following section, we will verify experimentally that a robotic hand designed based on the design principle can change grasping forces to grasp an object in both strongly and gently way, and can implement some representative manipulations of human hand in practice.

V. IMPLEMENTATION AND TEST OF THE MINI X-HAND

A. Implementation of the Mini X-Hand

According to the dual-actuated design principle, the final mechanisms and actuators (motor for fingers: Maxon DCX10L, gear GPX10 16:1, and encoder ENX10; motor for thumb: Maxon DCX10S, gear GPX10 256:1, and encoder ENX10) are both integrated in compact palm [see Fig. 3(a)], with critical mechanism parameters listed in Table II. The primary motion

$$\begin{bmatrix} \mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} - \mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o}\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\mathbf{G}\mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} & -\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o}\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\bar{\mathbf{G}}\mathbf{tr}_{\bar{\sigma}_0}^{\bar{\mathbf{f}}_h^o} \\ -\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\mathbf{G}\mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} & \mathbf{tr}_{\bar{\sigma}_0}^{\bar{\mathbf{f}}_h^o} - \mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\bar{\mathbf{G}}\mathbf{tr}_{\bar{\sigma}_0}^{\bar{\mathbf{f}}_h^o} \end{bmatrix}$$

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transmission relationship $\mathbf{U}_{\text{primary}}^{\text{robot}} \in \Re^{8\times 2}$ between the eight mechanical joints and two actuating sliders is

$$\mathbf{U}_{\text{primary}}^{\text{robot}} = \begin{bmatrix} \frac{1}{{}^1r_1\,{}^1r'_2+{}^1r_2\,{}^1r'_1}\begin{bmatrix} {}^1r'_2 \\ {}^1r'_1 \end{bmatrix} & & \\ & \ddots & \\ & & \frac{1}{{}^4r_1\,{}^4r'_2+{}^4r_2\,{}^4r'_1}\begin{bmatrix} {}^4r'_2 \\ {}^4r'_1 \end{bmatrix} \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 1-d_{12}/d_{14} & d_{12}/d_{14} \\ 1-d_{13}/d_{14} & d_{13}/d_{14} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.0830 & 0 \\ 0.1609 & 0 \\ 0.0653 & 0.0405 \\ 0.0929 & 0.0577 \\ 0.0193 & 0.0781 \\ 0.0257 & 0.1042 \\ 0 & 0.1051 \\ 0 & 0.1274 \end{bmatrix}. \tag{57}$$

The motion transmission relationship $\mathbf{U}_{\text{primary}}^{\text{human}} \in \Re^{8\times 2}$ obtained from human motion data (human data are extracted from [37], which consists of 30 subjects and 33 grasping types, and $\begin{bmatrix} c_1 \\ & c_4 \end{bmatrix}^{-1}$ is determined as $\begin{bmatrix} 0.0809 & \\ & 0.1026 \end{bmatrix}$) is

$$\mathbf{U}_{\text{primary}}^{\text{human}} = \begin{bmatrix} \frac{{}^1\mathbf{p}_1}{{}^1p_{11}} & & \\ & \ddots & \\ & & \frac{{}^4\mathbf{p}_1}{{}^4p_{11}} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \end{bmatrix} \cdot \begin{bmatrix} u_{11} & u_{12} \\ u_{41} & u_{42} \end{bmatrix}^{-1} \begin{bmatrix} c_1 & \\ & c_4 \end{bmatrix}^{-1} = \begin{bmatrix} 0.0809 & 0 \\ 0.1582 & 0 \\ 0.0622 & 0.0391 \\ 0.0914 & 0.0575 \\ 0.0186 & 0.0757 \\ 0.0261 & 0.1064 \\ 0 & 0.1026 \\ 0 & 0.1325 \end{bmatrix}. \tag{58}$$

The high proximity (see Fig. 6) and small differences ($\|\mathbf{U}_{\text{primary}}^{\text{robot}} - \mathbf{U}_{\text{primary}}^{\text{human}}\|_F^2 / \|\mathbf{U}_{\text{primary}}^{\text{human}}\|_F^2 = 0.0008$) between the two motion transmission matrices $\mathbf{U}_{\text{primary}}^{\text{robot}}$ and $\mathbf{U}_{\text{primary}}^{\text{human}}$ indicate that the designed robotic hand can, no doubt, replicate human grasping characteristics in practice.

Besides that, torque limits of the drag axes related to four fingers are 0.5 kgf·cm (ZN-YZX-011, FISHEYE, Guangdong, China; the product information can be found in https://item.taobao.com/item.htm?spm = a1z10.3-c-s.w4002-24001166676.11.5e2a6faeyT6vPa&id = 541204422700), torque limit of the drag axis related to thumb is 6 kgf·cm (ZN-YZX-033, FISHEYE, Guangdong, China; the product information can be found in https://item.taobao.com/item.htm?spm = a1z10.3-c-s.w4002-24001166676.79.7f166faefyTkZd&id = 541539124989), and the helical pitches of the lead screws are 2.44 mm/round. The designed mini X-hand can wear the human-sized data glove (CyberGlove III) to carry out experiments (see Fig. 8), indicating its compactness. In addition, Fig. 3(b) indicates that the mini X-hand can move all joints in fingers during natural motions, and possess the self-adaptive characteristics similar to a human hand. Fig. 3(c) demonstrates that the thumb design ensures the abilities of thumb-finger opposition and stepless adjustment.

To visualize several typical self-adaptive grasping processes, we illustrate three examples [as shown in Fig. 7(a)] and use Vicon system (position accuracy: 0.017 mm) to record the changes of four marker positions on four proximal phalanges (used for calculating angles of the four MCP joints), the displacement of

$$\begin{bmatrix} \delta\mathbf{Q}_{\text{passive}}(t) \\ \delta\bar{\mathbf{Q}}_{\text{passive}}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{Tr}_{\mathbf{x}}\mathbf{tr}_{\sigma_0}^{\mathbf{x}} + \mathbf{Tr}_{\sigma*}\mathbf{Tr}_{\mathbf{c}} \begin{bmatrix} \Gamma_1\left(\mathbf{F}_1^{\text{slider}}(t), \delta\sigma_0(t)\right) \\ \Gamma_2\left(\mathbf{F}_2^{\text{slider}}(t), \delta\sigma_0(t)\right) \end{bmatrix} & \mathbf{0} & \mathbf{Tr}_{\mathbf{x}}\mathbf{tr}_{\mathbf{u}}^{\mathbf{x}} & \mathbf{Tr}_{\mathbf{x}}\mathbf{tr}_{\mathbf{f}_h^o}^{\mathbf{x}} & \mathbf{0} \\ \mathbf{0} & \mathbf{Tr}_{\bar{\mathbf{x}}}tr_{\bar{\sigma}_0}^{\bar{\mathbf{x}}} & \mathbf{Tr}_{\bar{\mathbf{x}}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{x}}} & \mathbf{0} & \mathbf{Tr}_{\bar{\mathbf{x}}}\mathbf{tr}_{\bar{\mathbf{f}}_h^o}^{\bar{\mathbf{x}}} \end{bmatrix} \begin{bmatrix} \delta\sigma_0(t) \\ \delta\bar{\sigma}_0(t) \\ \delta\mathbf{u}(t) \\ \delta\mathbf{f}_{ht}^o \\ \delta\bar{\mathbf{f}}_{ht}^o \end{bmatrix} \tag{55}$$

$$\begin{bmatrix} \delta\mathbf{u}(t) \\ \delta\mathbf{f}_{ht}^o \\ \delta\bar{\mathbf{f}}_{ht}^o \end{bmatrix} = \begin{bmatrix} -\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\mathbf{G}\mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} & -\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\bar{\mathbf{G}}\mathbf{tr}_{\bar{\sigma}_0}^{\bar{\mathbf{f}}_h^o} & -\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} \\ \mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} - \mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o}\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\mathbf{G}\mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} & -\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o}\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\bar{\mathbf{G}}\mathbf{tr}_{\bar{\sigma}_0}^{\bar{\mathbf{f}}_h^o} & \\ -\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o}\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} & & & \\ -\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\mathbf{G}\mathbf{tr}_{\sigma_0}^{\mathbf{f}_h^o} & \mathbf{tr}_{\bar{\sigma}_0}^{\bar{\mathbf{f}}_h^o} - \mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} {}^o\bar{\mathbf{G}}\mathbf{tr}_{\bar{\sigma}_0}^{\bar{\mathbf{f}}_h^o} & \\ -\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\left({}^o\mathbf{G}\mathbf{tr}_{\mathbf{u}}^{\mathbf{f}_h^o} + {}^o\bar{\mathbf{G}}\mathbf{tr}_{\mathbf{u}}^{\bar{\mathbf{f}}_h^o}\right)^{-1} & & & \end{bmatrix} \times \begin{bmatrix} \delta\sigma_0(t) \\ \delta\bar{\sigma}_0(t) \\ \delta\mathbf{w}^o(t) \end{bmatrix} \tag{56}$$

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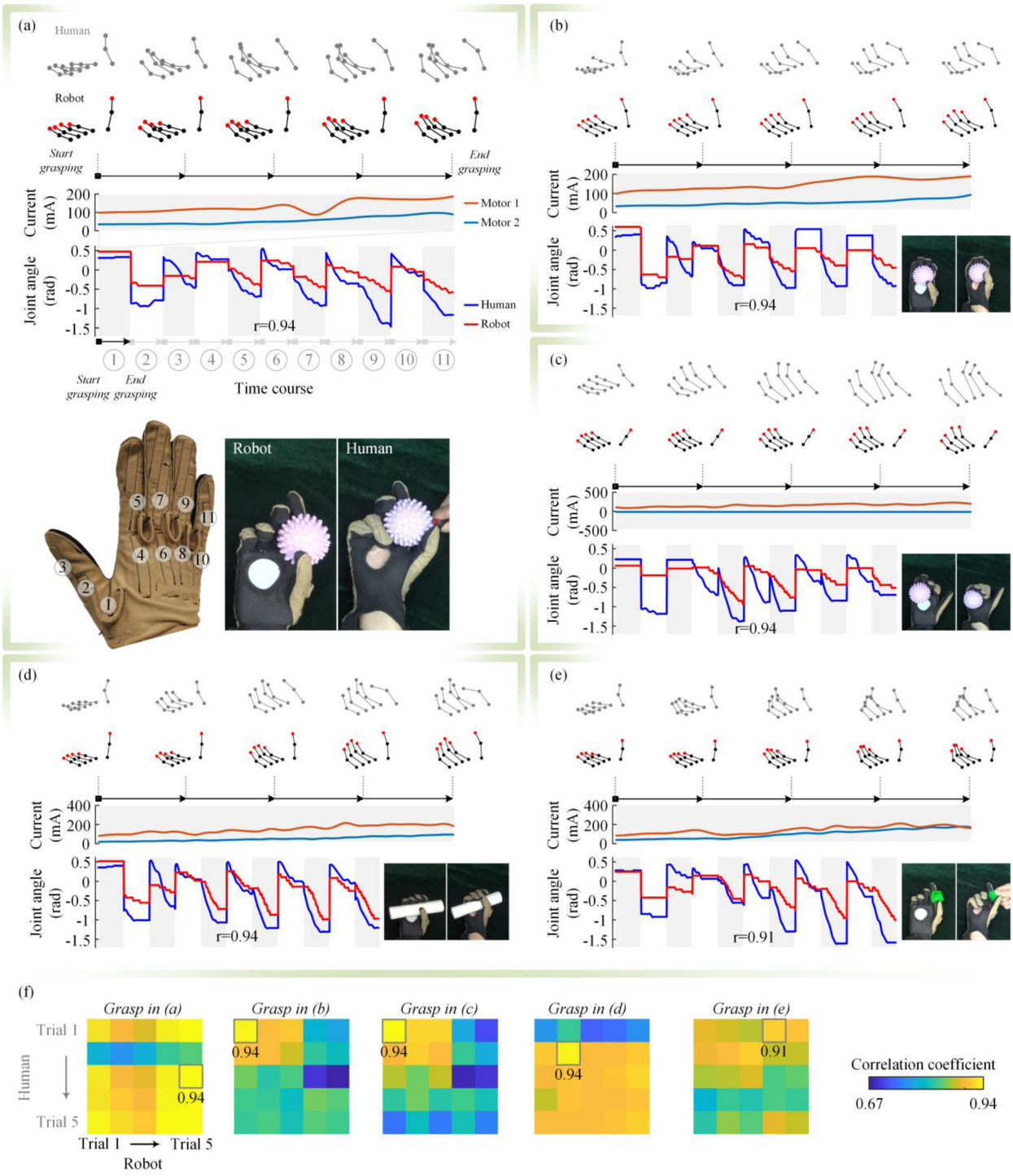


Fig. 8. Evaluation of anthropomorphic motion capability of the mini X-hand in several typical self-adaptive grasps. (a)–(e) Recorded joint angle data (corresponding postures are shown above) and current data in the five typical grasps. (f) Correlation between the recorded motion data of a human hand (five trials) and motion data of the robotic hand (five trials). There exist 25 correlation coefficients for one grasping type, and all values are more than 0.67 (the highest value is always more than 0.91), which demonstrates the well anthropomorphic motion capability of the robotic hand. Note that, (a)–(e) show the five trial pairs with the highest correlation coefficients (gray box) in (f).

the two sliders in these three typical adaptive grasping behaviors through two markers on sliders and three marker positions on hand (used for deriving a hand coordinate system). From the angles of the four MCP joints, we can derive the deformation of the four springs, and then predict the angles of all eight joints of the four fingers of the mini X-hand based on (3) [as shown in Fig. 7(a)]. Since we have obtained the deformation processes of the springs and the differential process of the differential sliders in the whole grasping process, we can obtain the adaptive behavior curves of the three grasping processes (the dynamic

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process of the curve with time refers to the supplementary video, and final curves are shown in the three curves in Fig. 7(b); the starting process represented by $\min\{\sigma_1^*(t), \sigma_2^*(t)\} < 2$ mm is ignored considering that the small value of the $\sigma_0(t)$ would cause high fluctuations of the curve). These three curves show three significantly different trajectories, indicating three typical adaptive behavior characteristics of the mechanism.

B. Evaluation of Anthropomorphic Motion Capability

In order to verify the anthropomorphic motion capability of the robotic hand during self-adaptively grasping objects in different grasping types, we make the robotic hand and a human hand wear a data glove (CyberGlove III) to grasp a ball (with thumb, index, and middle fingers; with all digits; with index, middle, and ring fingers), a cylinder and a small square. The test of the human subject is approved by the Chinese Ethics Committee of Registering Clinical Trials. The participant gave has written informed consent. These five representative grasps cover different shapes and sizes of objects, significantly different self-adaptive ways and significantly different grasping types. The data glove records angle data of 11 joints as shown in Fig. 8(a) (sampling frequency: 20 Hz). Each grasp executes five times. At the same time, the current data of the two motors are recorded.

The data extracted from the glove sensors are first truncated to reserve the interval between start grasping and end grasping (i.e., the flat portions of the curves are delated). Then, the reserved data are fitted by a cubic spline curve and 101 resampled data in equal intervals of time are obtained. We splice all the joint data together [as shown in Fig. 8(a)–(e)], and calculate the correlation between the motion data of a human hand and motion data of the robotic hand (101×11 samples). The results are shown in Fig. 8(f). It can be found that the highest correlation coefficients in the five grasps are all more than 0.91, indicating a high degree of similarity between human motion and robotic motion. Fig. 8(a)–(e) shows the five groups of curves with the highest correlation and the corresponding final postures. The high motion correlations and the final anthropomorphic postures prove that the design principle can ensure that the robot hand maintains anthropomorphic grasping behaviors during the typical self-adaptive grasping processes.

C. Test of Grasping and Manipulation Abilities

We evaluate the daily grasping capabilities of the mini X-hand, utilizing Feix taxonomy [30] as the testing condition. Feix taxonomy [30] contains 33 grasping types, which cover almost all human grasps in daily life. In prior work [3], the four-actuated X-hand can complete 30 grasping types, and the unimplemented grasping types (distal type, tripod variation, and adduction grip) owe to the limitation in dexterous manipulation and abduction/adduction movements of the MCP joints of the middle and index fingers. As shown in Fig. 9(a), despite the number of actuators in the mini X-hand is reduced to 2 and control of thumb abduction/adduction is changed to manual adjustment, it can still achieve 29 grasping types in the 30 grasping types that the X-hand can complete. The unique failure grasp is ring grasp. Ring grasp cannot be achieved because the mini X-hand cannot realize actively medial-to-center flexion of fingers. Only one motor is used to drive the four fingers.

It can be further found that despite the abundant grasping types achieved by the mini X-hand, these grasps can be further classified in terms of virtual fingers (VF) opposed to thumb (see Fig. 10). According to the VF, five thumb positions (five classes) are enough to address these grasps. They include: thumb abducted; thumb opposed to lateral side of index finger; thumb opposed to tip of index finger; thumb opposed to middle part between index and middle fingers; thumb opposed to middle part between middle and ring fingers. This means that users do not need to adjust thumb positions each time in practical use. If two adjacent grasps fall in the same class (one of the five classes in Fig. 10), thumb position could be held still. More exactly, if the frequency of occurrence of the jth grasp ($j = 1, \ldots, 29$) in daily use is known as Oc_j, the probability that users do not need to adjust thumb position in practical use can be calculated as $p = (Oc_1 + \cdots + Oc_5)^2 + (Oc_6 + \cdots + Oc_8)^2 + (Oc_9 + \cdots + Oc_{11})^2 + (Oc_{12} + \cdots + Oc_{16})^2 + (Oc_{17} + \cdots + Oc_{29})^2$ supposing each grasp is an independent event. The Yale human grasping dataset, recording prehensile human hand use during the daily work activities of two housekeepers and two machinists, provides use frequencies of the 29 grasps in daily life [38], [39]. According to the data, the probability that users do not need to adjust thumb position can be calculated as $p = 31\%$, which means that about one out of every three grasps may need no thumb position adjustment in practical use.

Due to the dependence relationship between the geometric orientation of the hand and grasping forces, readers may doubt that whether the hand can grasp the same object with different grasping forces, i.e., grasp an object in both strongly and gently way. This is vital for practical use in daily life. The theoretical derivation in the last section has demonstrated that the contact forces can be adjusted by changing the two motion inputs of motors. Here, we further carry out a practical experiment. We make the mini X-hand grasp a foam ball embedded with a pinchmeter (Pinchmeter P200, Biometrics Ltd., Newport, U.K.) in two motor states (power ON and power OFF), as shown in Fig. 11. The current thresholds of motors are set as 200 mA, and the sample frequency of the pinchmeter is 100 Hz. In the "power ON" state, the motors of the hand are given positive position commands by a computer, until the motor currents reach the thresholds, which means that the hand has grasped the ball under the current condition. In the "power OFF" state, which means that the motor is powered OFF, due to the existence of lead screw, the sliders can be maintained at the positions (return back a small distance due to the mechanism clearance). The hand can still grasp the ball, but with lighter gripping forces (see Fig. 11). The different gripping forces in different motor states mean that the mini X-hand is able to stably grasp a specific object with different grasping forces. Note that, motor currents are limited here. The pinchmeter that is mainly used to test the pinch force is not used to test the maximum gripping force here. We will test it using a dynamometer to be introduced in the following.

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Fig. 9. Test of grasping and manipulation capabilities of the mini X-hand. (a) Hand can complete 29 types of daily grasps in Feix taxonomy. (b) Hand can manipulate a tweezer and a ball (in-hand manipulation examples) and screw bottle cap. More grasping and manipulation details refer to the supplementary video.

On the aspect of manipulation, prior neurological research [40] indicated that intrinsic manipulative hand movements of human fall into the following three classes.

1) Simple synergies (single manipulated pattern; flexors of thumb and fingers work together, extensors of thumb and fingers work together).
2) Reciprocal synergies (single manipulated pattern; thumb and fingers show reciprocal movements, such as flexion of the fingers with adduction or extension of the thumb).
3) Sequential patterns that need dexterous interdigit movements (some digits support an object, whereas other digits need reposition during manipulation).

A high proportion of intrinsic movements fall into the first two classes [40]. Due to 1) independent module design of the thumb and four fingers and 2) bidirectional actuating characteristics, the mini X-hand has the capability to simultaneously flex thumb and fingers to manipulate a tweezer [see Fig. 9(b)], or control reciprocal movements of thumb and fingers to manipulate a ball [see Fig. 9(b)]. Conversely, due to the lack of electrically driven abduction/adduction function of thumb, the hand cannot achieve manipulations that need the participation of abduction/adduction function of thumb (part of the second class). Due to the existence of only one motor for the four fingers, the hand cannot independently control each finger so that it is not possible to achieve the reposition of digits if the object is in the hand. However, if the manipulated object is half fixed outside the hand, the hand still has the ability to readjust positions of thumb and finger to achieve dexterous manipulation of the object, e.g., unscrewing the cap on a water bottle [see Fig. 9(b)]. The realization of these manipulations (also refer to the supplementary video) proves the manipulative capabilities of the dual-actuated principle. The dual-actuated principle endows the mini X-hand with the

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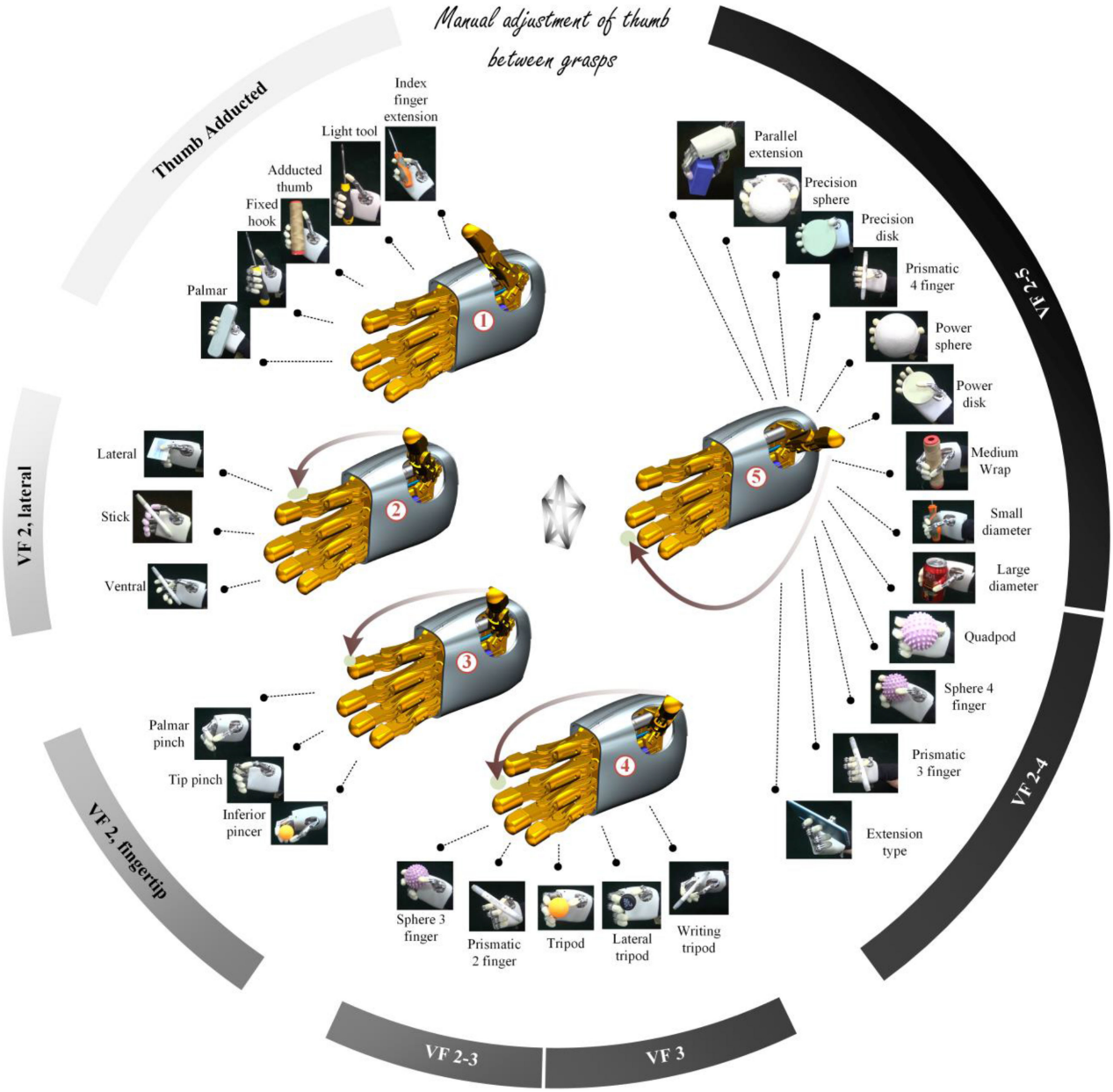


Fig. 10. Five thumb positions are able to achieve the 29 daily grasps as well as manipulations. As there is no overly precise position required for these five thumb positions, position switching is simple and quick.

capability of achieving a limited portion of manipulation capabilities of human hand.

In addition, this article mainly focuses on the dual-actuated design principle for replicating human grasping characteristics and manipulation abilities, the motors are mostly inherited from the X-hand in our prior work [3]. The maximum holding force of the mini X-hand during grasping a dynamometer (Dynamometer G200, Biometrics Ltd., Newport, U.K.) is 8.82 N (refer to Fig. 12), due to the choices of the motor power (only 3 W here) and torque limits of the drag axes. The gripping forces can be improved through choosing a different motor with greater motor power (such as an 8 W brushless Maxon motor, which has the similar sizes to the used 3 W brush motor). Since motor power equals to motor speed multiplying motor torque, if speeds are same, the maximum output torque of the later motor would be no doubt higher. If the maximum output torque of the motor becomes larger, drag axes with higher torque limits can be chosen (the reason has been discussed in Section IV-D, paragraph 3), so that grasping forces can be further improved. Since motor torque or motor power usually has relations with motor sizes that relate to hand sizes, in design of a robotic hand to address real-life scenarios, it is needed to synthetically consider the motor power, motor sizes, and the slip torque of drag axes according to practical requirements, and sometimes, compromises among the three aspects (motor power, motor sizes, and the slip torque of drag axes) are necessary.

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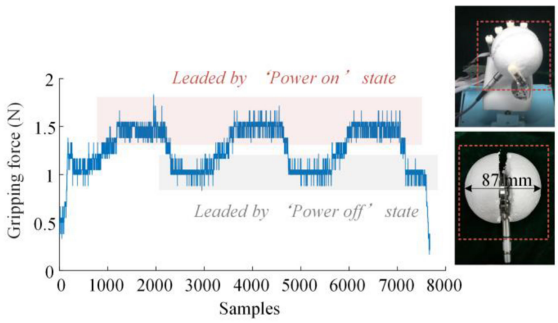


Fig. 11. Gripping force variation test of the mini X-hand. The ball is initially put near the base of fingers and thumb, as shown above. An experimenter holds the pinchmeter to prevent the ball from falling down at the beginning. Once the ball is grasped, the experimenter removes his hand, and there is no human interference during the experimental procedure.

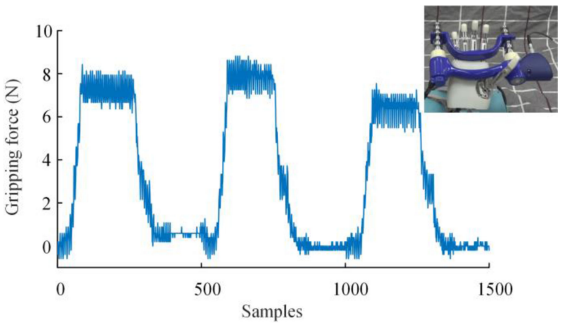


Fig. 12. Gripping force test of the mini X-hand using a dynamometer. The dynamometer is initially put near the base of fingers and thumb, as shown above. The fingers and thumb are required to move simultaneously. There is no human interference during the experimental procedure.

VI. Conclusion

This article aimed to address one core challenge in the multi-fingered robotic hand design—establishing a design principle of the five-fingered robotic hand for reproducing human sizes, motions and grasping/manipulation functions with extremely few actuators—and proposed a new dual-actuated design principle. At a high level, we constructed a design theory allowing thumb and fingers independently driven and four fingers synergistically driven so that the designed hand can carry inherent movement characteristics of human hands, which underlie abundant grasping functions and dexterous manipulation abilities. At a low level, a novel single-actuated differential mechanism that can reproduce two eigengrasps, which can account for 97% human grasping characteristics among four fingers, was created so that the designed robotic hand can implement both the anthropomorphic motions before contacting objects and self-adaptively pinching/enveloping objects of arbitrary shapes like a human hand. The quasi-static analysis was further carried out to clarify the determination factors of free and passive movements of the dual-actuated robotic hand, grasping force controllability and manipulability of the dual-actuated robotic hand. Finally, we designed a practical hand named mini X-hand and carried out experimental tests. Experimental results demonstrated that the mini X-hand possesses anthropomorphic sizes as shown by wearing a human-sized data glove, anthropomorphic motion capability (high correlation between the motion data of a human hand and motion data of the robotic hand), the ability to adapt to the shape of objects, the ability to stably grasping an object with different grasping forces, abundant daily grasping functions of 29 daily grasps compiled in Feix taxonomy, and dexterous manipulation abilities like manipulating a ball (in-hand manipulation) and screwing bottle cap. These results mean that the designed robotic hands based on the proposed design principle would be close to a human hand on both aspects of functionality and anthropomorphism. Such closeness implies that robotic hands based on such a design principle could be applied as useful tools in a wide range of areas, such as end-effectors in service robots, general grippers in automatic factories, remote end-effectors in dangerous environments, and prosthetic hands for amputees.

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